CURRENT TOPICS IN COMPUTER SCIENCE



Business Intelligence Systems and Analytics

BUSINESS INTELLIGENCE AND FUNDAMENTAL OF VISUALIZATION

Trong Nhan Phan, PhD

OUTLINE

- Business Intelligence Systems
- Fundamental of Visualization
- Summary
- References

BUSINESS INTELLIGENCE SYSTEMS

BI FEATURES

- BI task
  - Providing decision support for specific goals defined in the context of business activities in different domain areas.
- BI foundation
  - BI decision support mainly relies on empirical information based on data.
- BI realization
  - The decision support has to be realized as a system using the actual capabilities in information and communication technologies (ICT).
- BI delivery
  - A BI system has to deliver information at the right time to the right people in an appropriate form.

Wilfried Grossmann Stefanie Rinderle-Ma, Fundamentals of Business Intelligence, Springer, 2015.

BI CONTEXT

- By scenarios
  - Standardized reports for a dedicated part of the business
  - Performance monitoring
  - A feedback on strategy formulation
  - Strategic resource
- By perspective
  - Production
  - Customer
  - Organization
  - Etc.

GOALS OF BI

- Key Performance Indicator (KPI)
  - A KPI links the activities of the business to objectives by defining a measurable quantity.
- Influential factors
  - Attributes that may influence the behavior of the KPI in any BI perspective.
- Analytical goals
  - Descriptive: reporting, segmentation, interesting behaviro detection
  - Predictive: regression, classification

DISCUSSION



BI TASKS TO DO

WHAT WOULD WE DO?

- Sales dashboard
  - Sales forecast
- Delivery performance
- Reservation performance
- Customer dashboard
  - Repeat rate
- Feedback dashboard
  - Feedback score
- Item dashboard
  - Item ABC analysis
- Etc.

BI TASKS

- Data task
- Business and data understanding task
- Modeling task
- Analysis task
- Evaluation and reporting task

DATA TASK

- A prerequisite for all BI activities
- The main goal is to organize available information about the business and its environment
- Not only retrieve data but also reorganize and collect additional data for special purposes

BUSINESS AND DATA UNDERSTANDING TASK

- Determine BI context in business
- An initial formulation of goals and KPIs
- Data needed to achieve those goals

MODELING TASK

- It aims at setting up an analytical business model, i.e., a formal model that allows precise answers for the analytical goals, and data related models

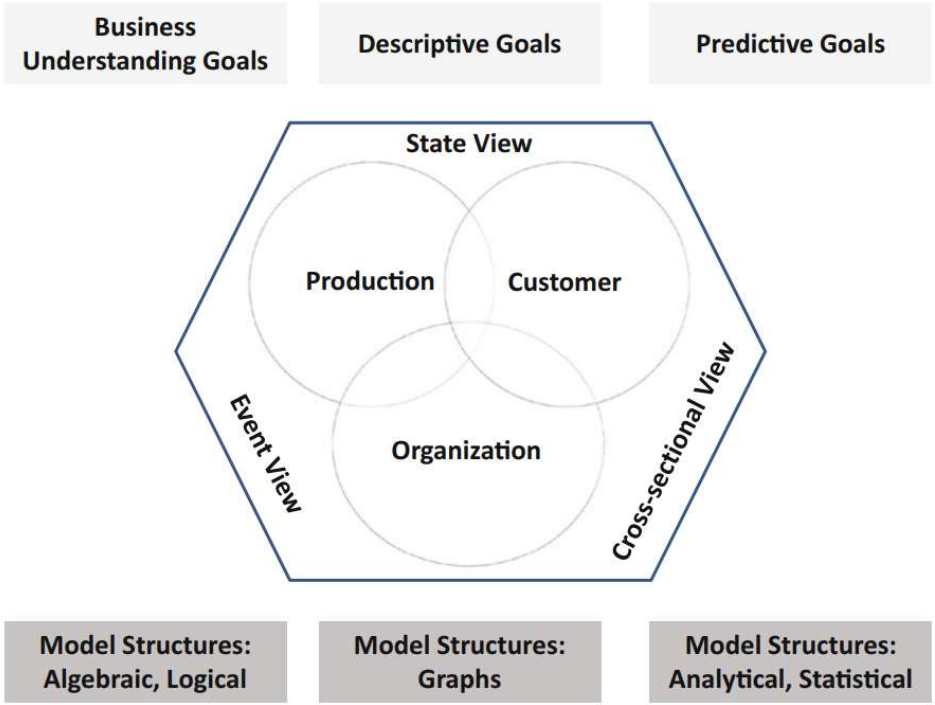


Fig. 1.2 Overview on modeling activities

Wilfried Grossmann Stefanie Rinderle-Ma, Fundamentals of Business Intelligence, Springer, 2015.

ANALYSIS TASK

- Algorithms to compute a solution for the analytical goal within the model.
  - Statistics methods
  - Data mining
  - Machine learning
  - Etc.

EVALUATION AND REPORTING TASK

- Evaluation task: view the analysis results from two different perspectives
  - The context of analytical goal
  - The context of business
- Reporting task: interprete the results to domain knowledge by description and visualization

THE iMine METHOD

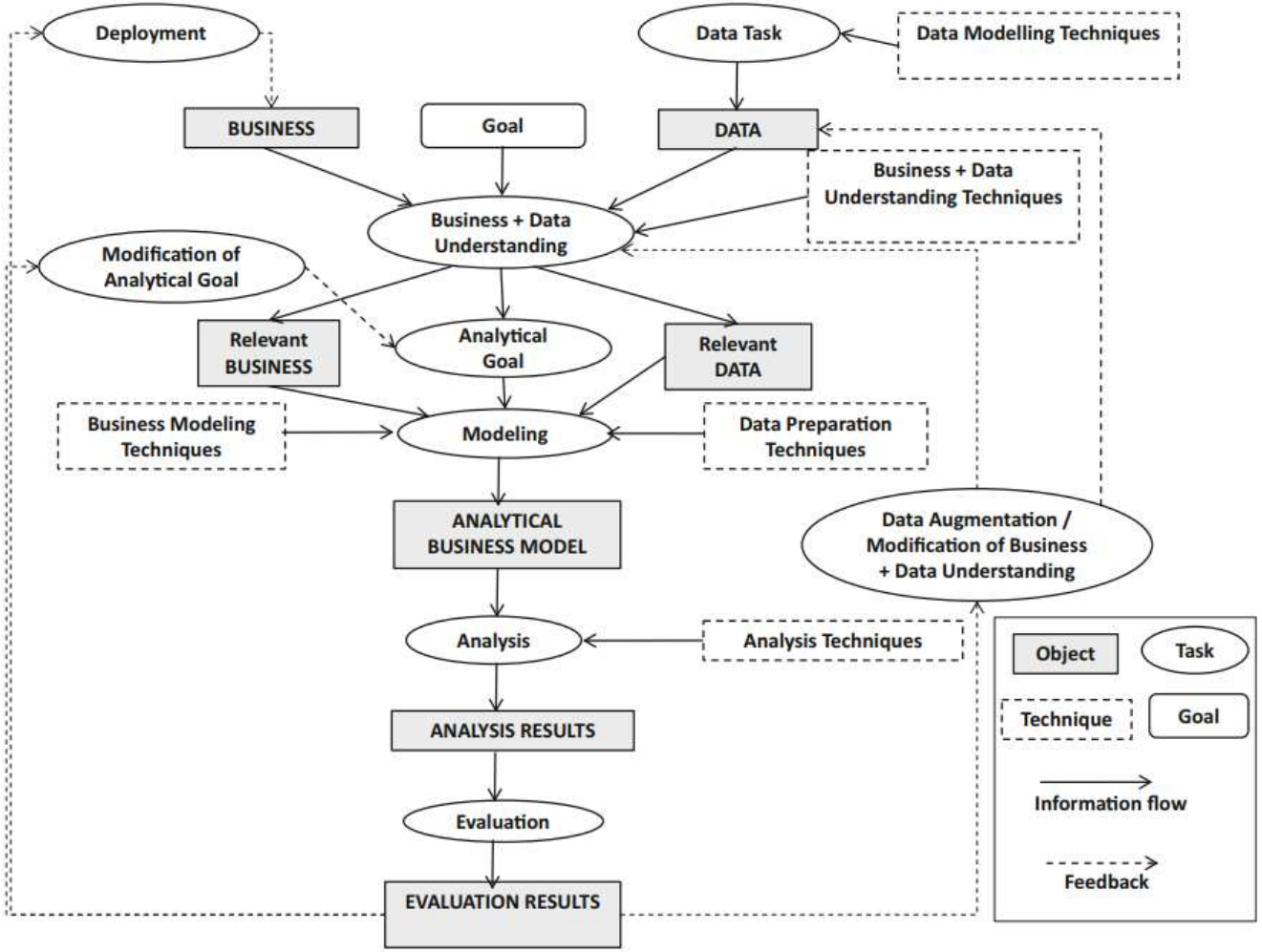


Fig. 1.4 The *iMine* method

Wilfried Grossmann Stefanie Rinderle-Ma, Fundamentals of Business Intelligence, Springer, 2015.

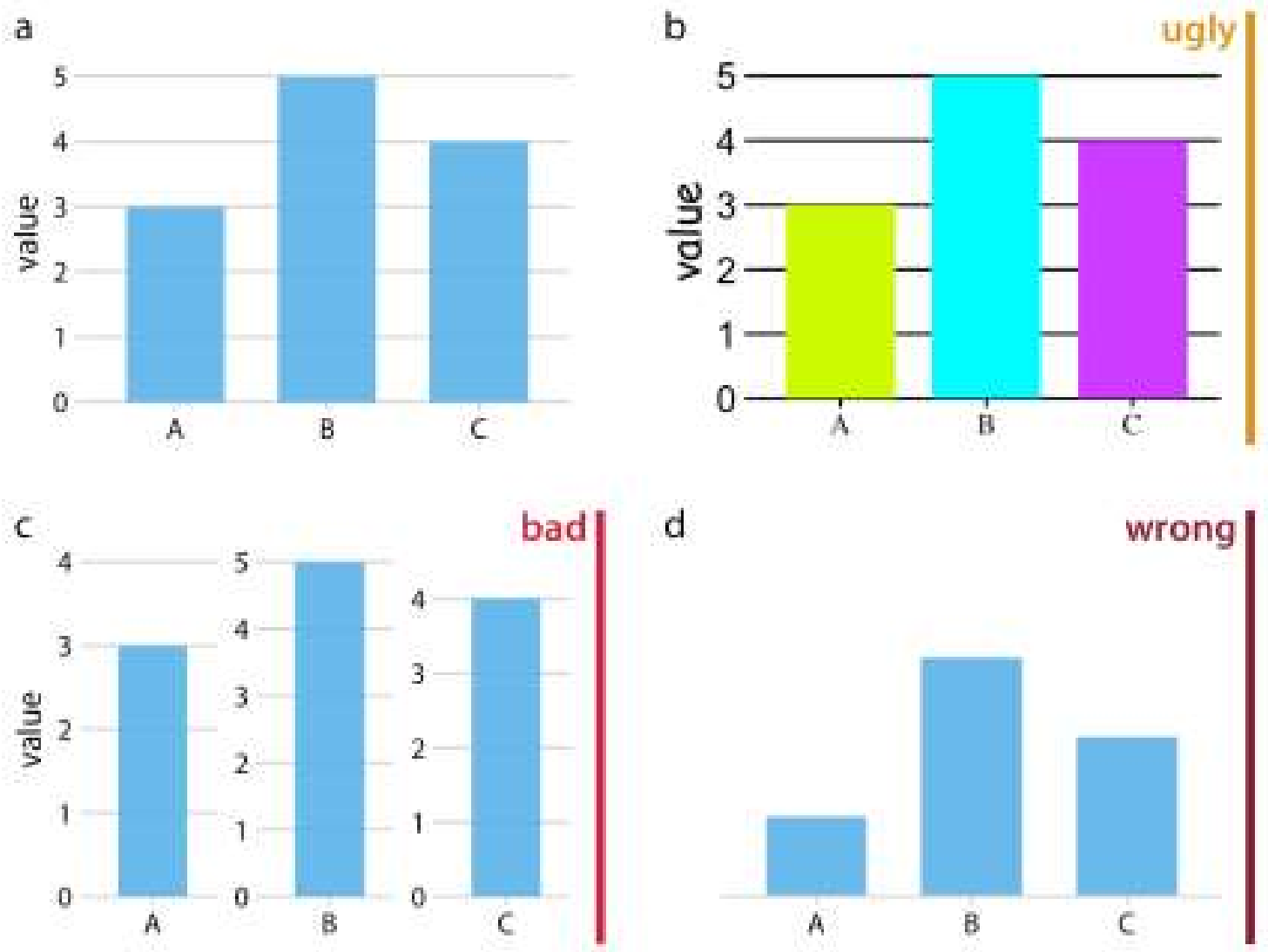
DISCUSSION



DELIVERY DASHBOARD

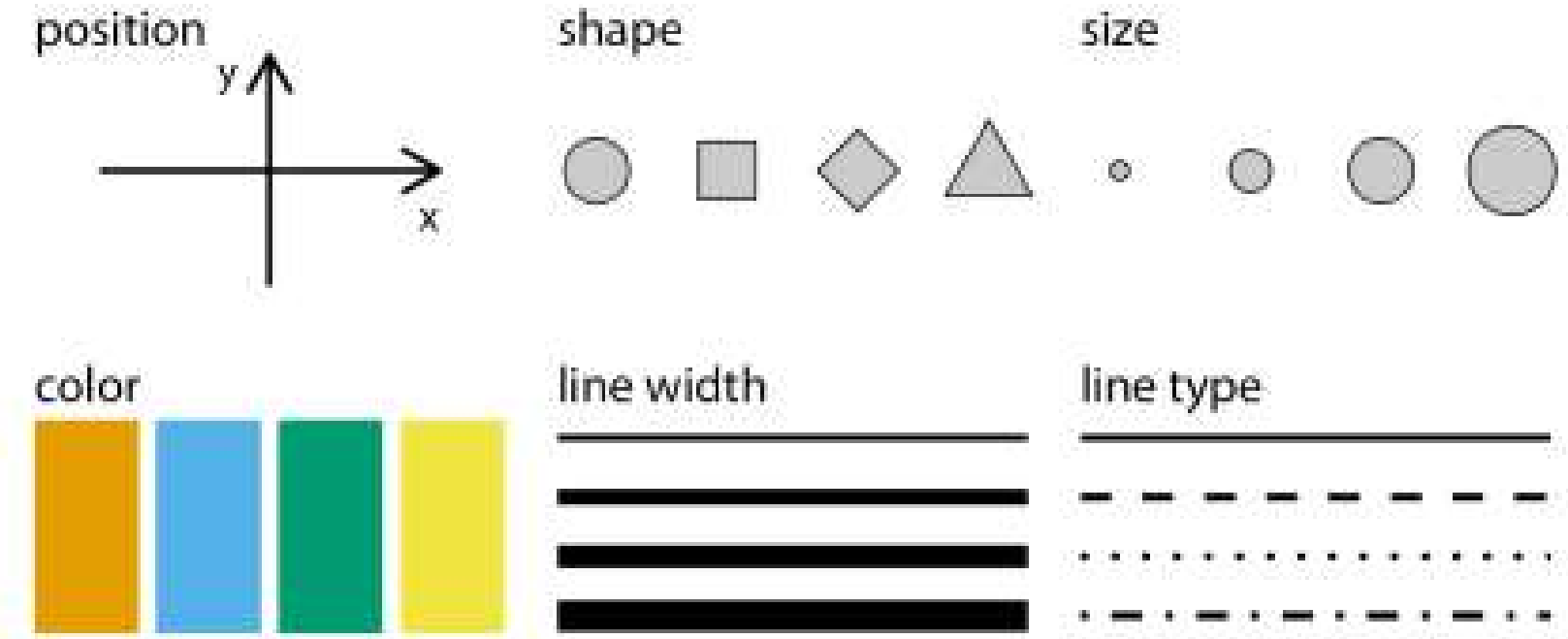
FUNDAMENTAL OF VISUALIZATION

CHART PROBLEMS



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

AESTHETICS IN DATA VISUALIZATION



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

TYPES OF DATA

Type of variable	Appropriate scale	Example
Quantitative, numerical continuous	Continuous	1.2, 5.8, 10, 3e-2
Quantitative, numerical discrete	Discrete	1, 5, 8, 12
Qualitative, categorical unordered	Discrete	Rose, Violet, Blossom
Qualitative, categorical ordered	Discrete	Bad, good, excellent
Datetime	Discrete or Continuous	2023-02-23 04:35:58
Text	None or Discrete	The cake is too burnt

DISCUSSION



DATA VISUALIZATION

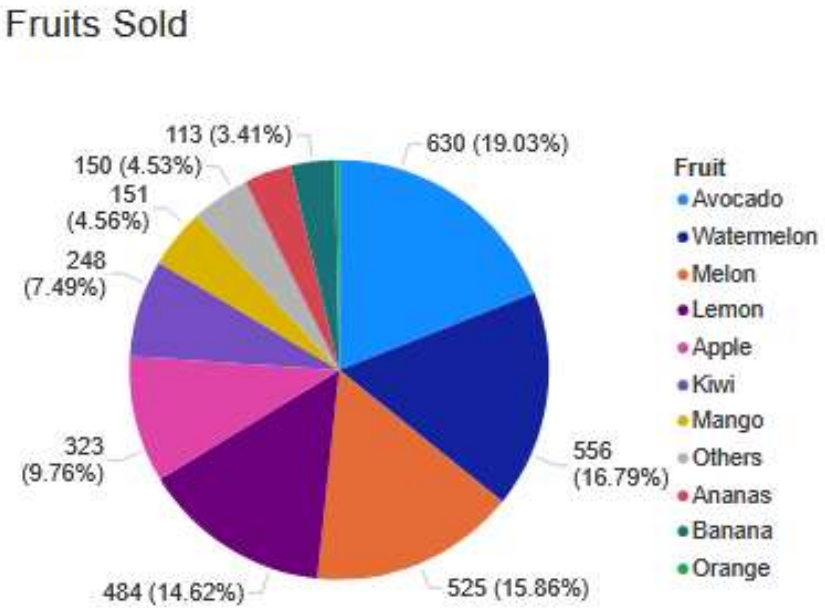
Month	Day	Location	Station ID	Temperature (°F)
Jan	1	Chicago	USW00014819	25.6
Jan	1	San Diego	USW00093107	55.2
Jan	1	Houston	USW00012918	53.9
Jan	1	Death Valley	USC00042319	51.0
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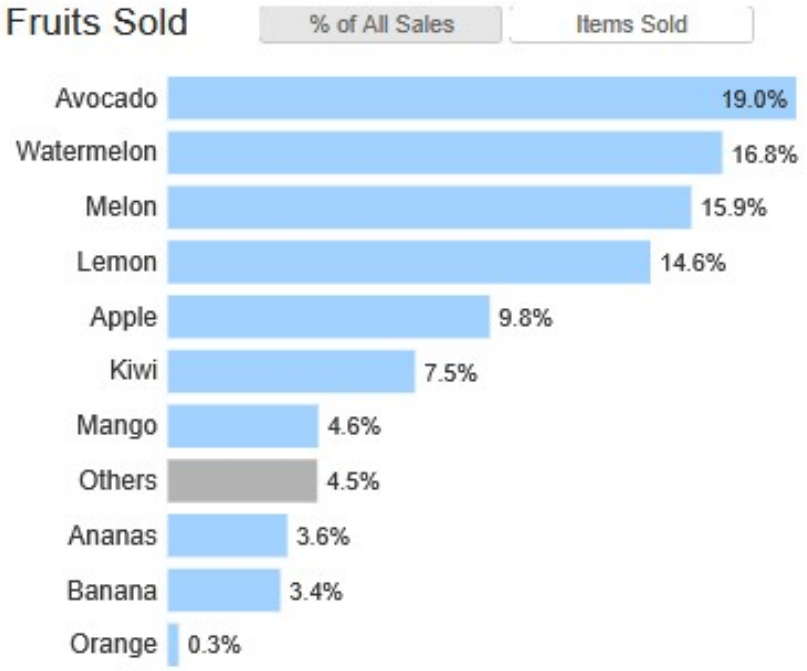
Rank	Title	Weekend gross
1	*Star Wars: The Last Jedi*	$71,565,498
2	*Jumanji: Welcome to the Jungle*	$36,169,328
3	*Pitch Perfect 3*	$19,928,525
4	*The Greatest Showman*	$8,805,843
5	*Ferdinand*	$7,316,746

Countr	Year	Status	Life expectancy
Belgium	2001	Developed	78
Lithuania	2000	Developed	71.6
Iceland	2013	Developed	82.4
Iceland	2012	Developed	82.5
Australia	2014	Developed	82.7
Cyprus	2009	Developed	79.3
Italy	2012	Developed	82
Italy	2011	Developed	82
Italy	2010	Developed	81.8
Netherlar	2013	Developed	81.4
Singapore	2011	Developed	82.2
Spain	2013	Developed	82.4
Switzerlar	2009	Developed	82.1
Switzerlar	2008	Developed	82
Austria	2012	Developed	88
Cyprus	2000	Developed	78.1
Iceland	2002	Developed	84
Ireland	2011	Developed	84
Japan	2003	Developed	81.9
Japan	2002	Developed	81.8
Luxembou	2012	Developed	81.1
New Zeala	2012	Developed	81.1
Norway	2006	Developed	84
Spain	2008	Developed	81.3
Sweden	2004	Developed	83
Austria	2007	Developed	81
Belgium	2011	Developed	83
Germany	2008	Developed	79.9
Luxembou	2008	Developed	80

Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

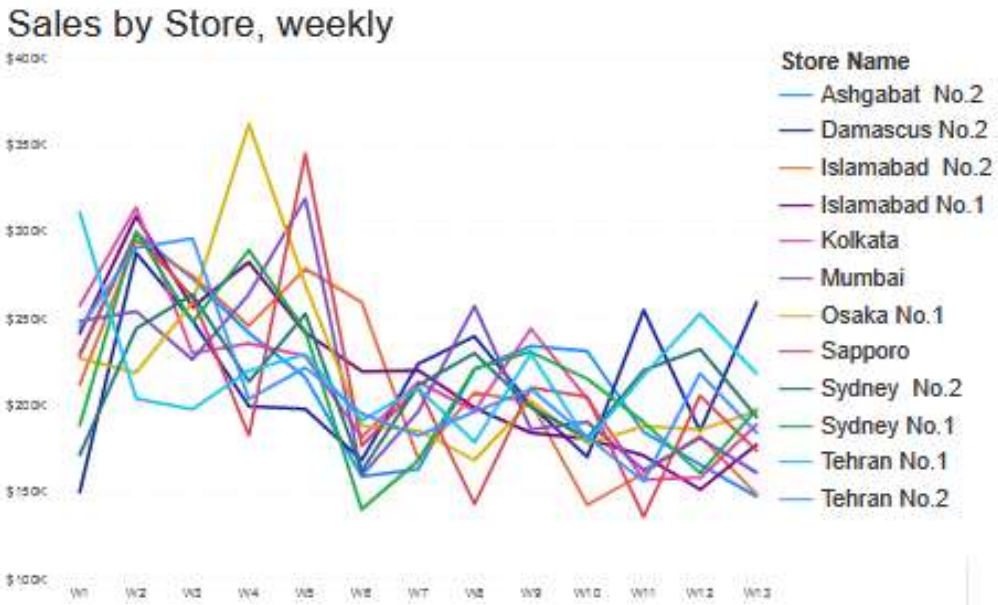
BAD VS. GOOD CHARTS





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BAD VS. GOOD CHARTS





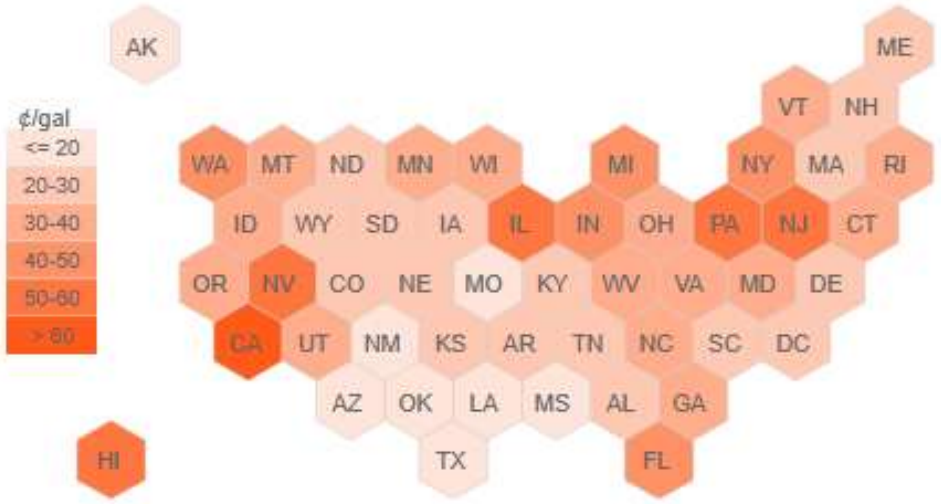
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BAD VS. GOOD CHARTS

Gasoline Tax by US State as of July 2021

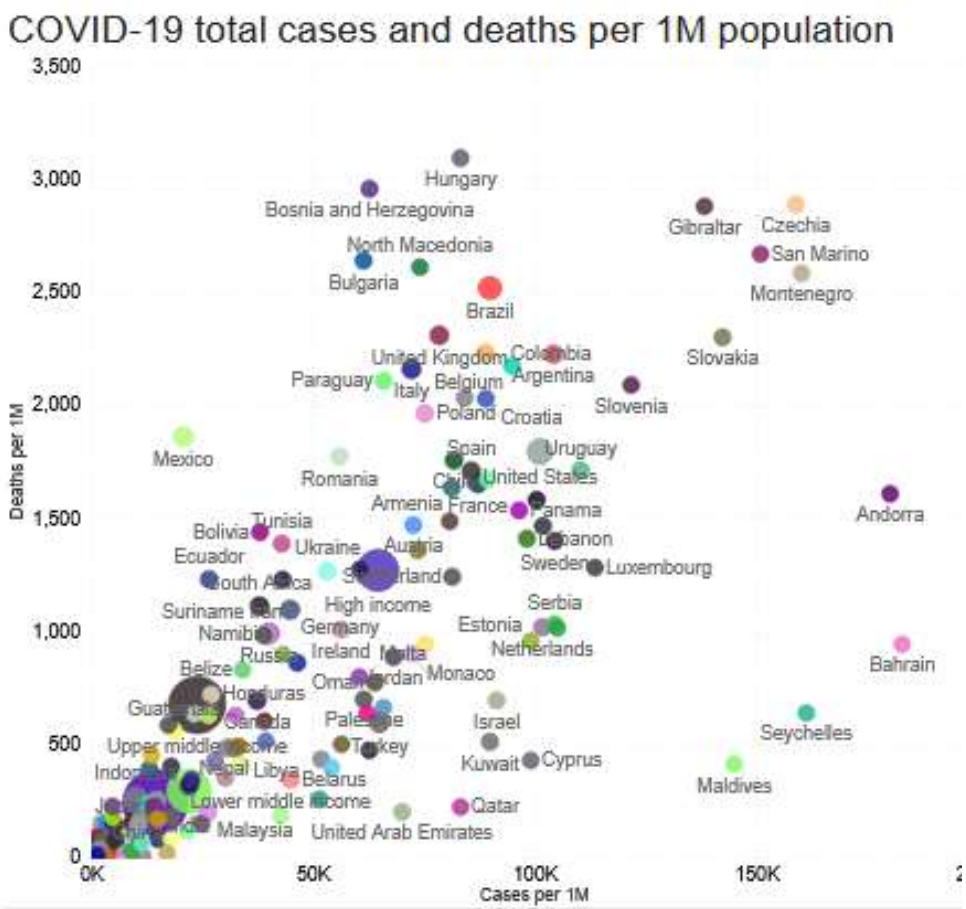


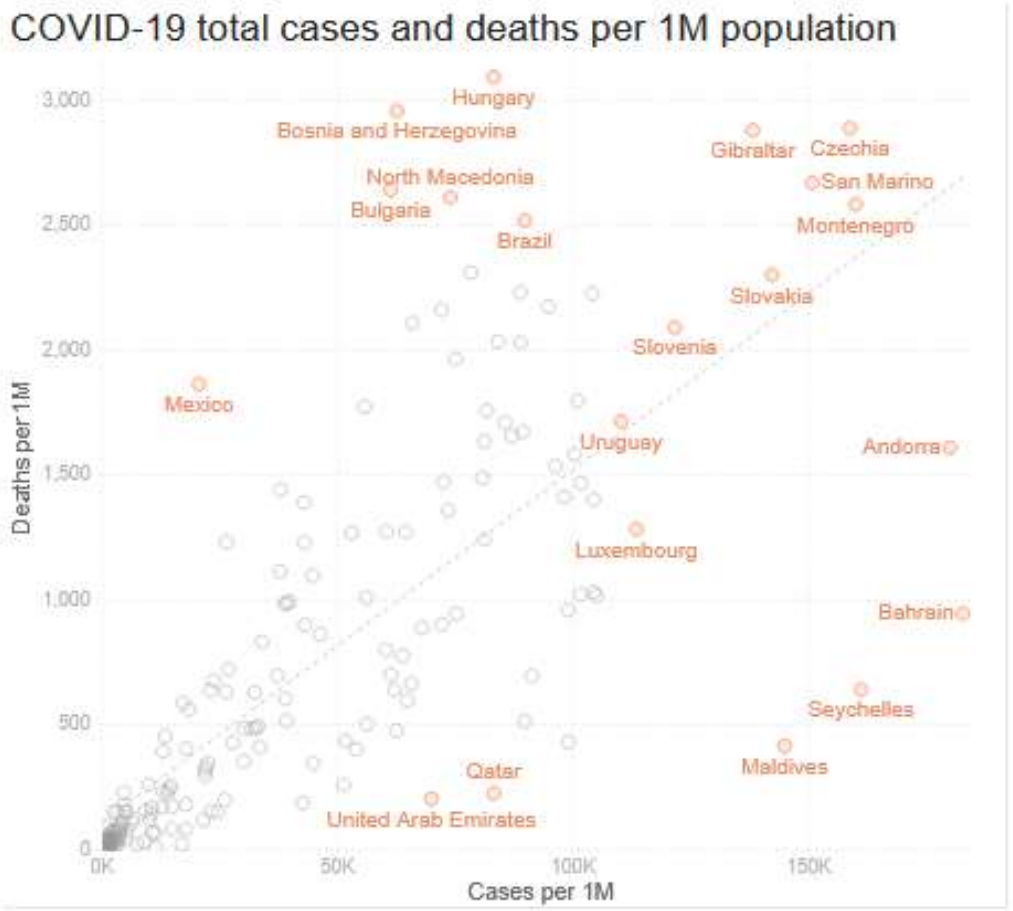
Gasoline Tax by US State as of July 2021



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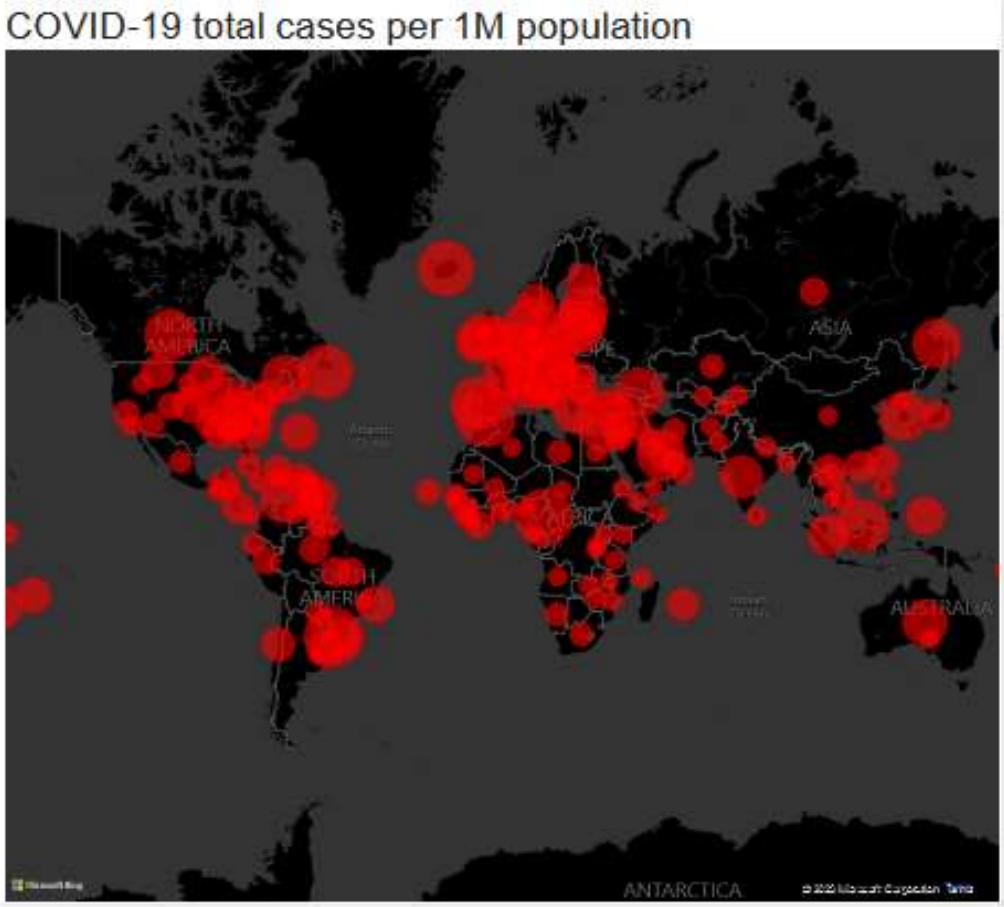
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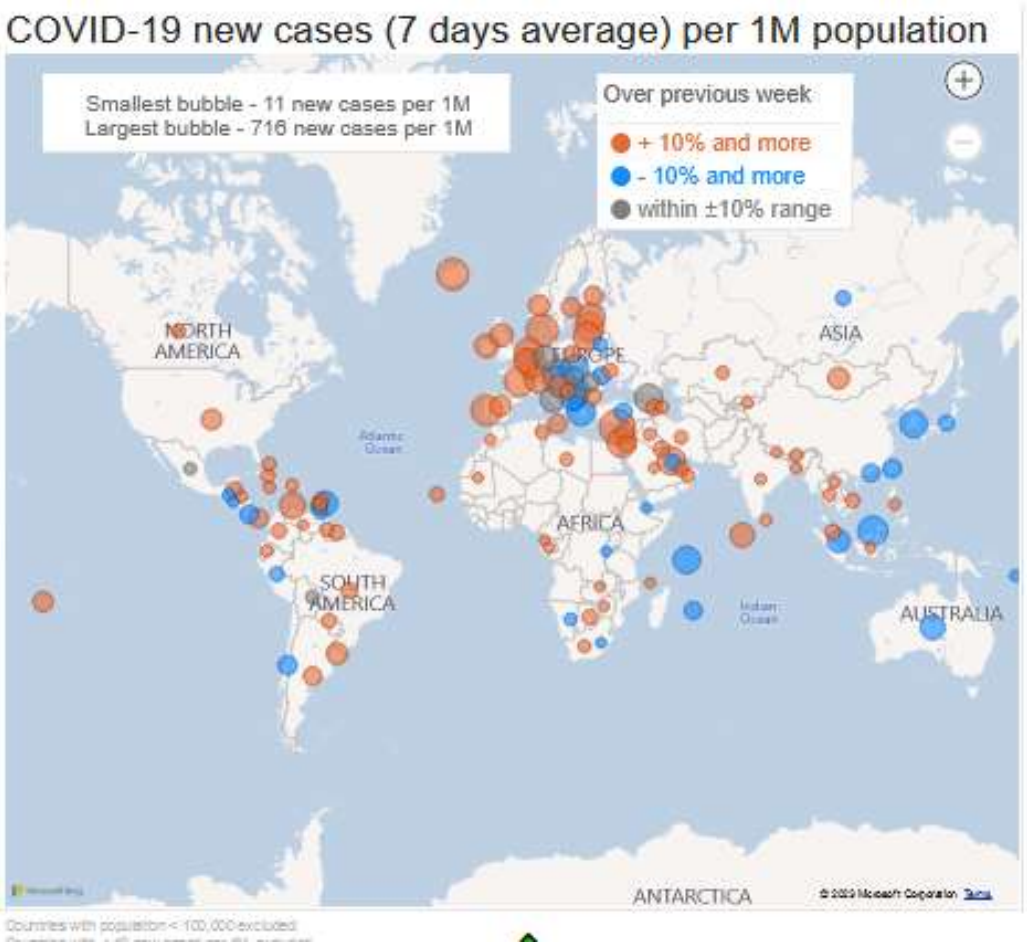




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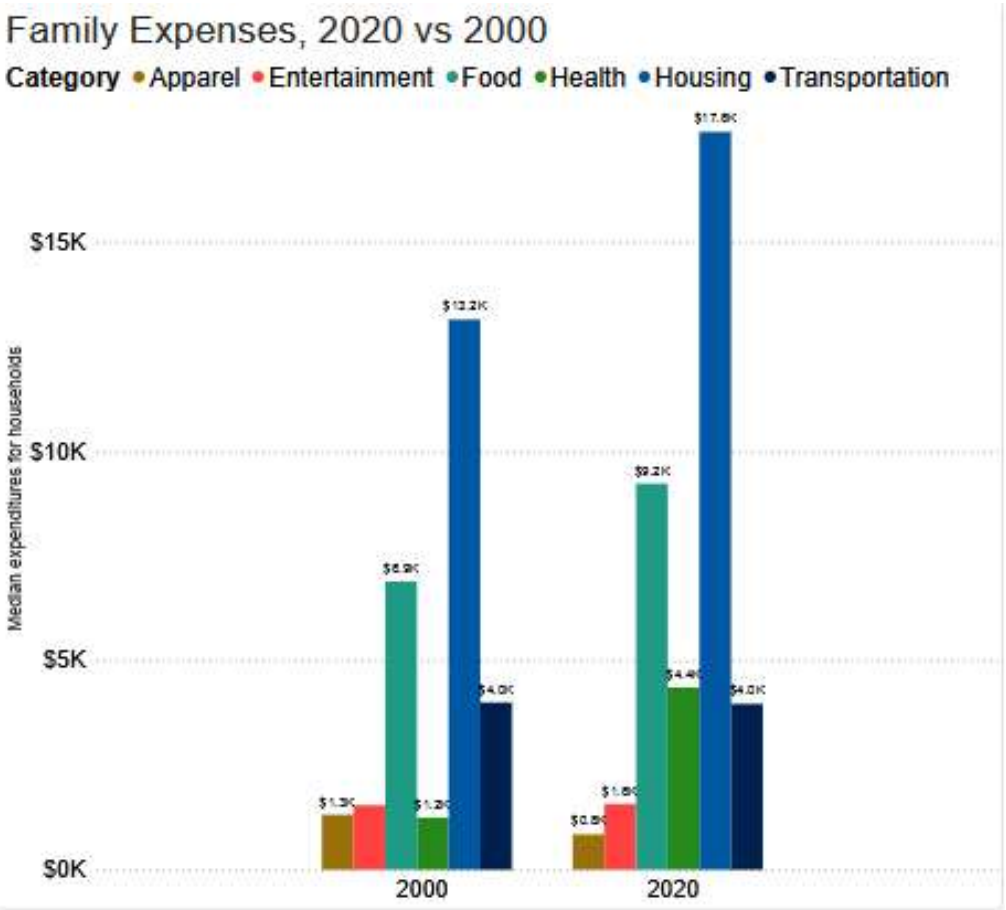
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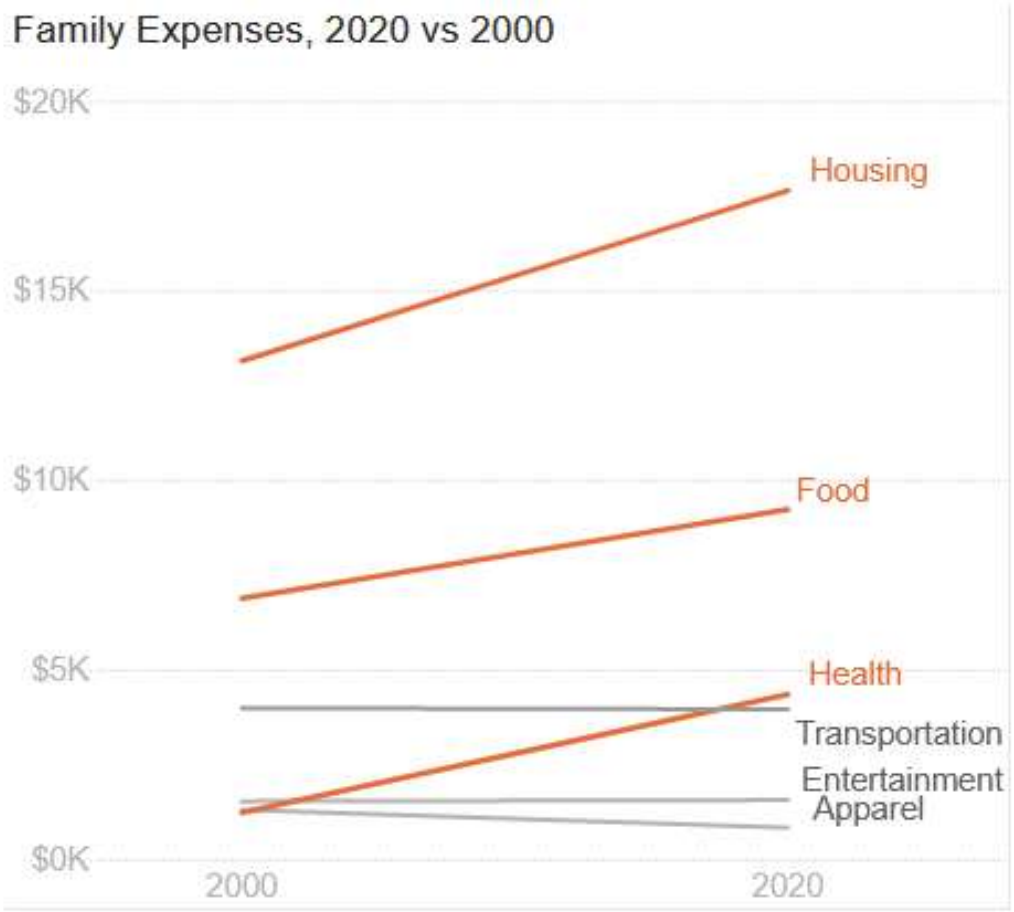




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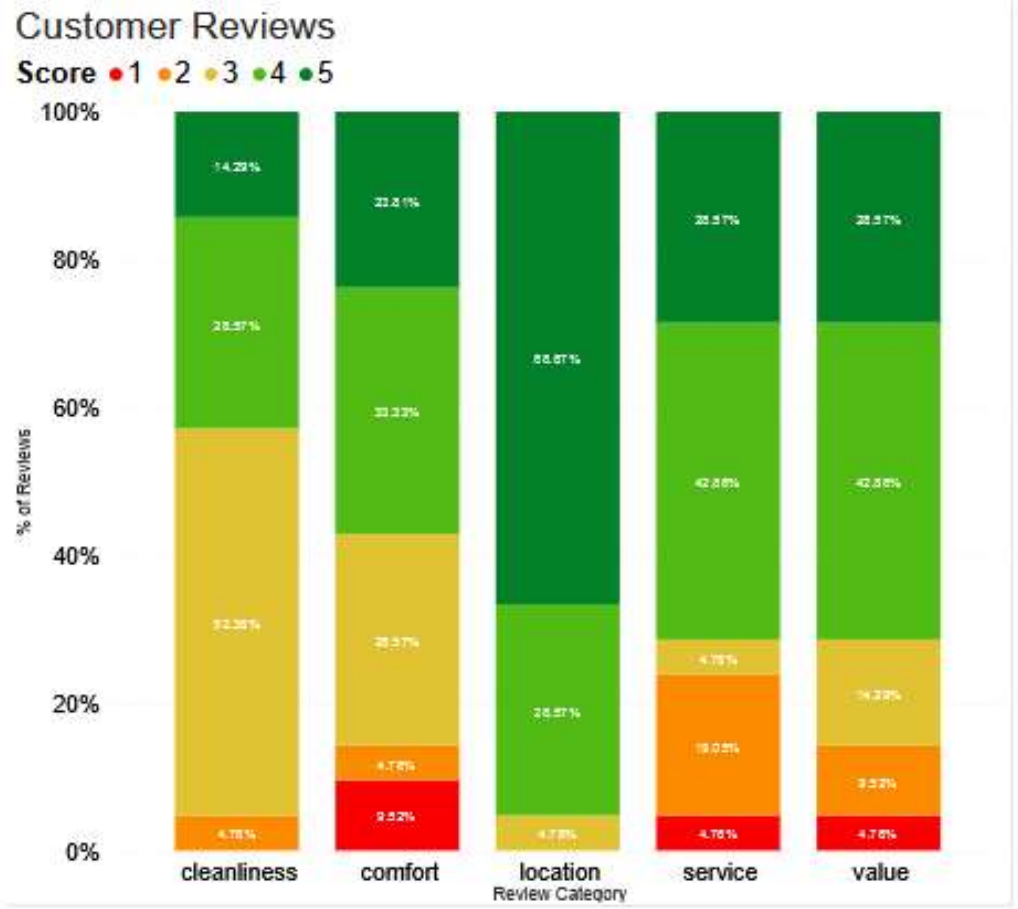
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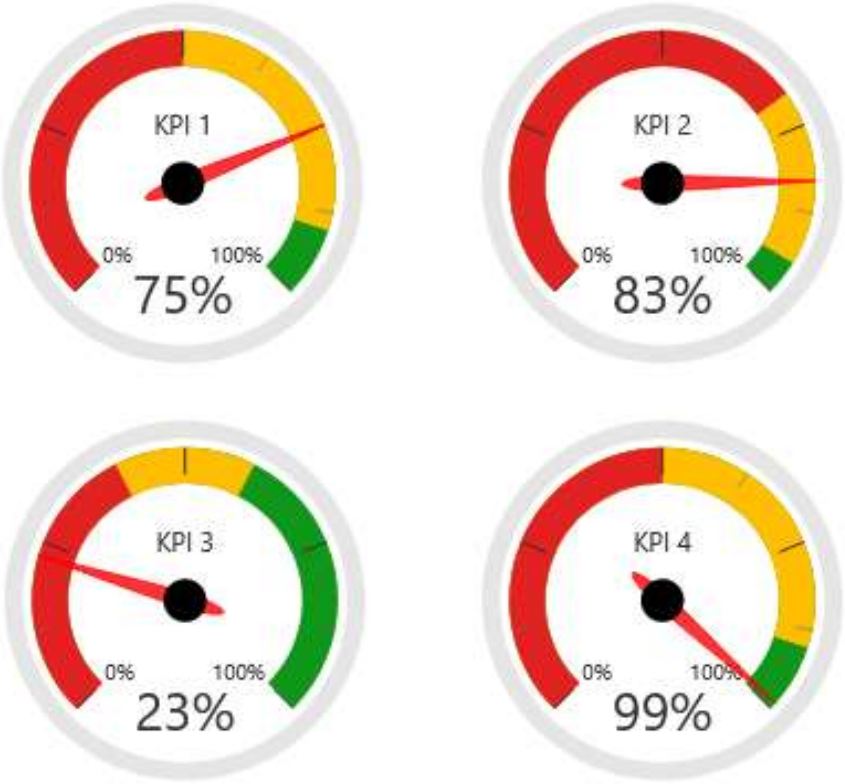
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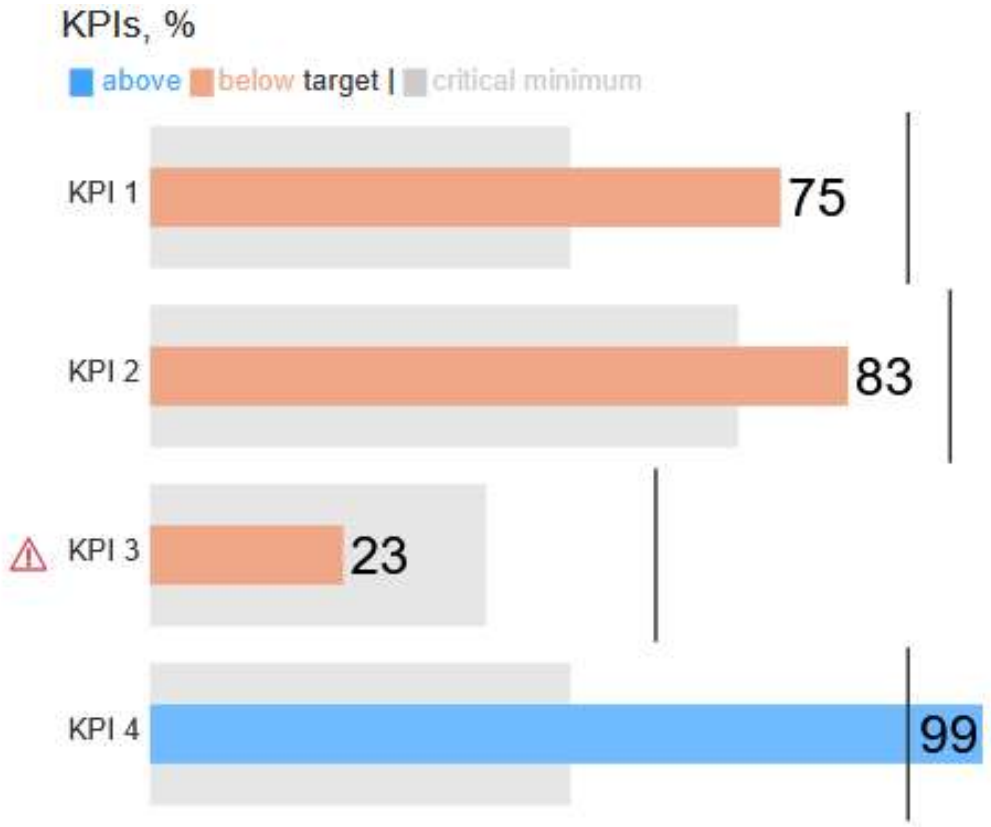




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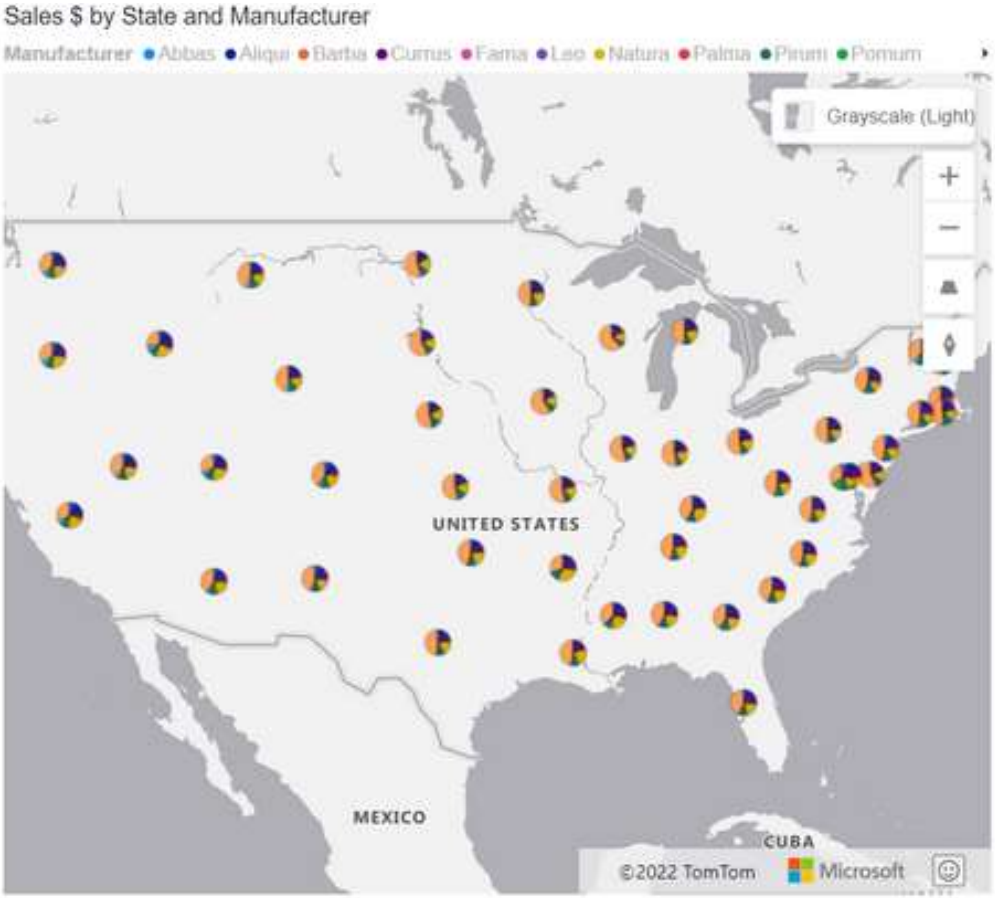
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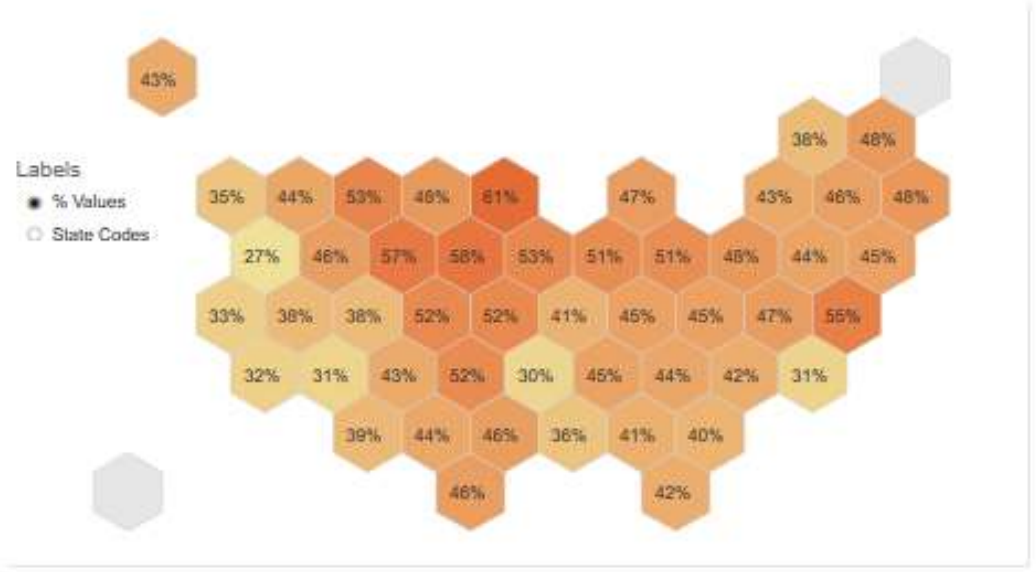


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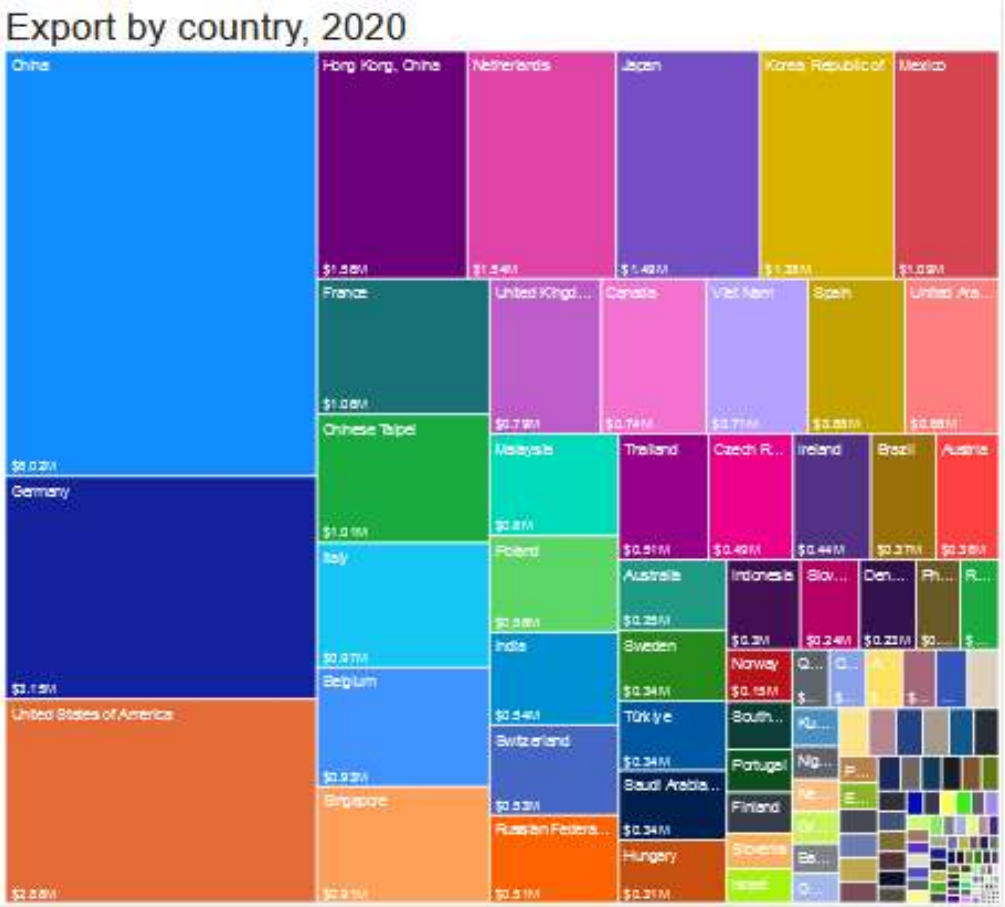


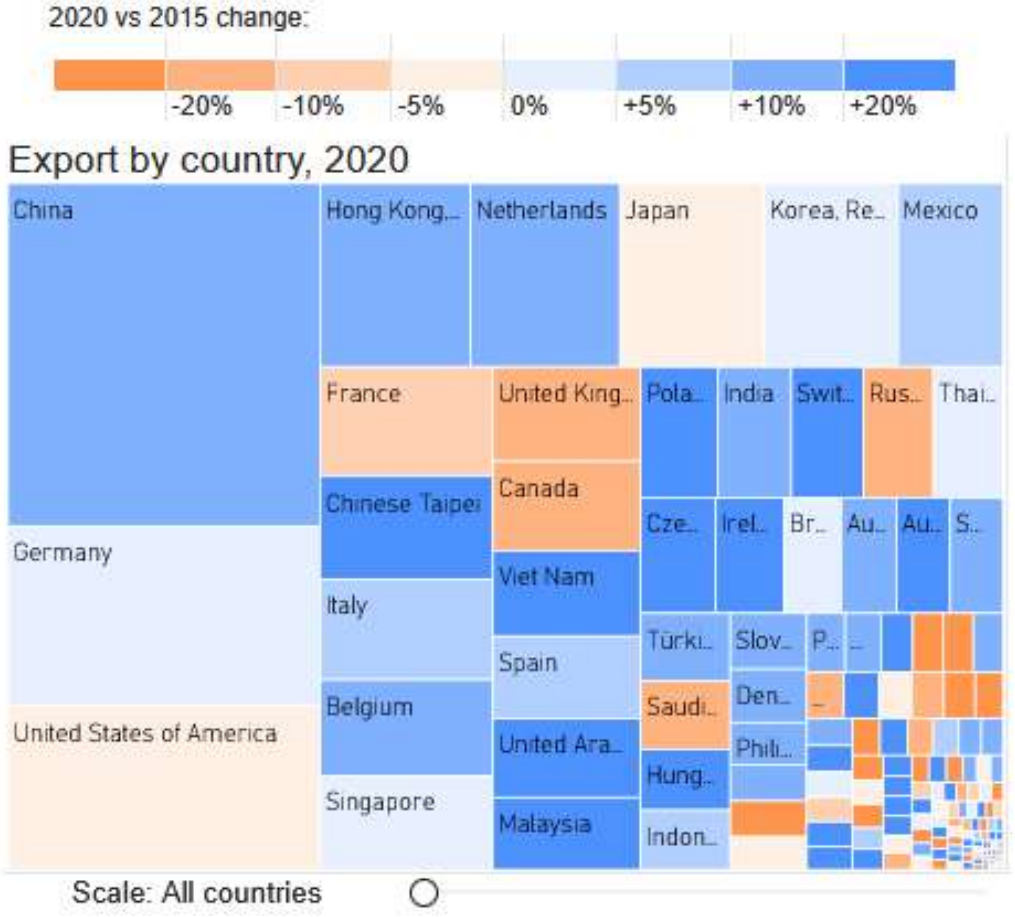




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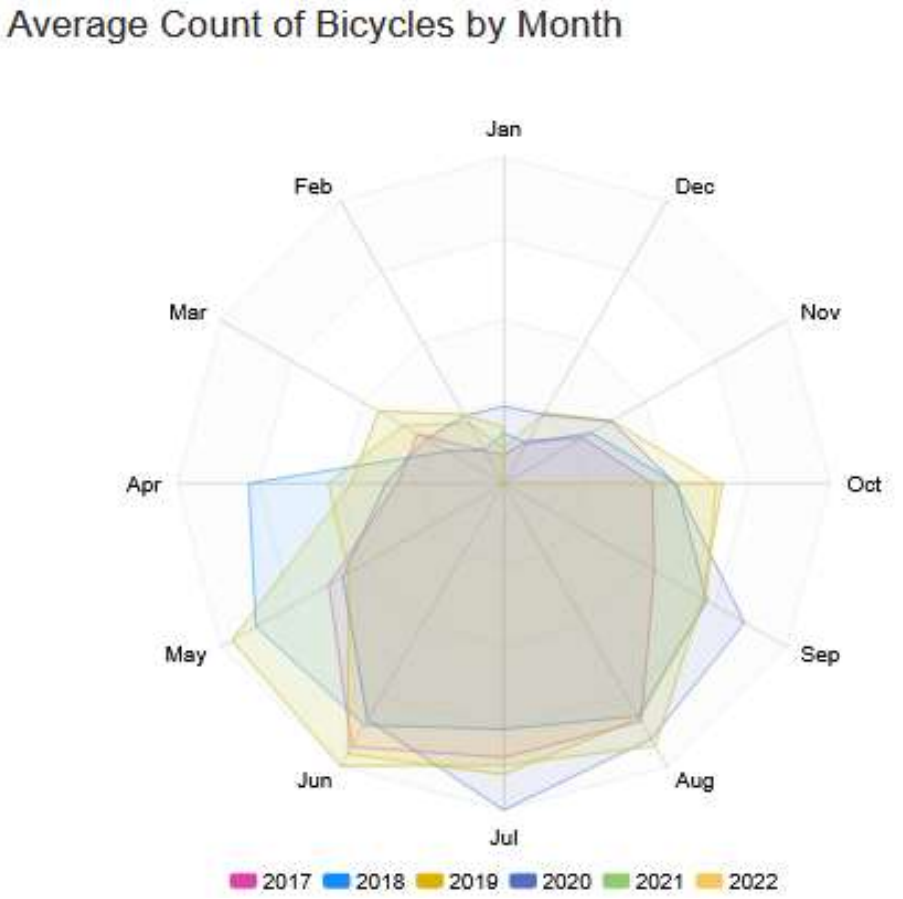
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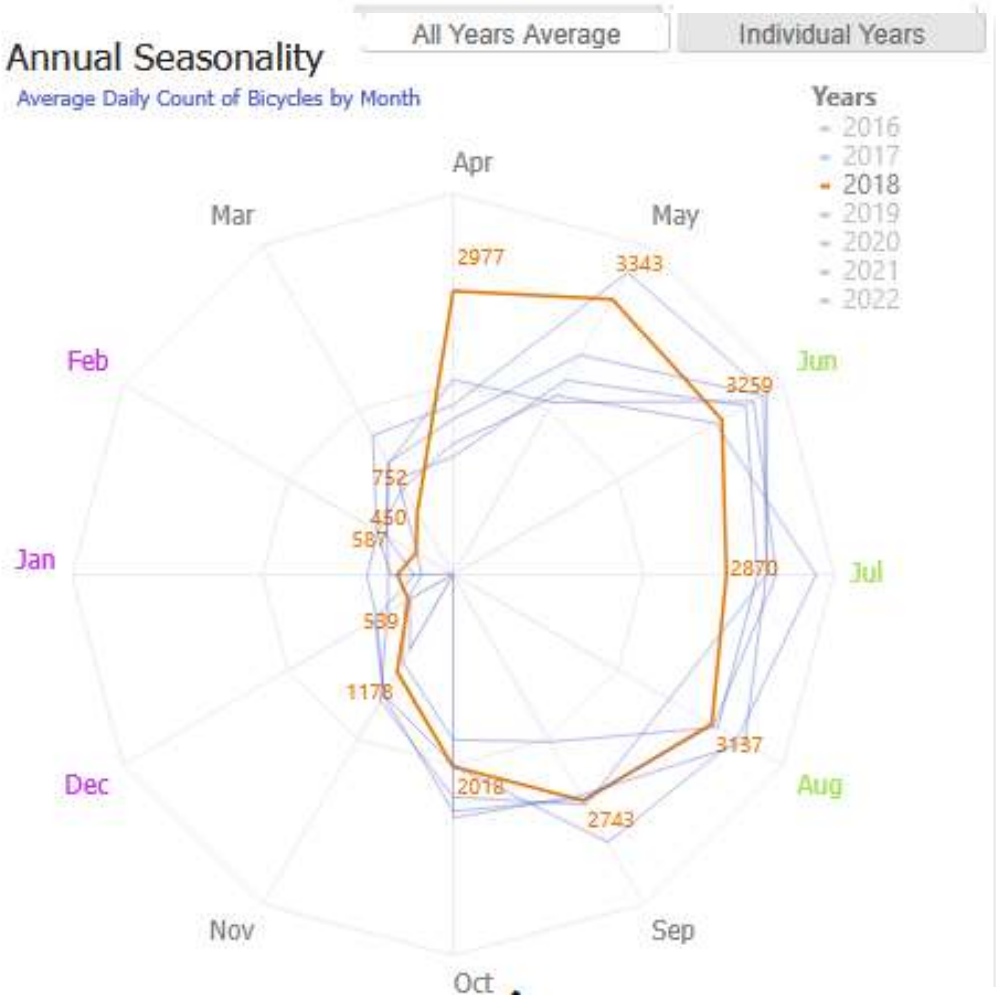




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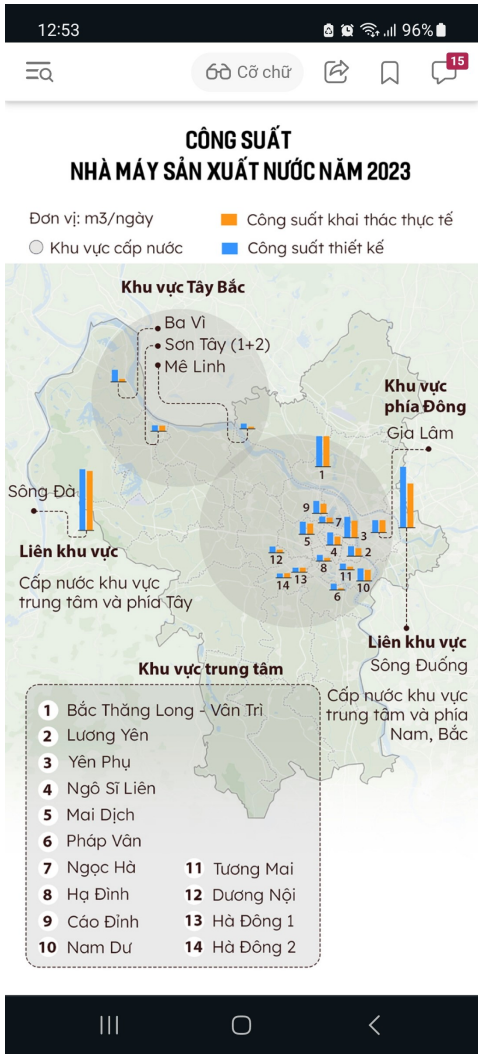
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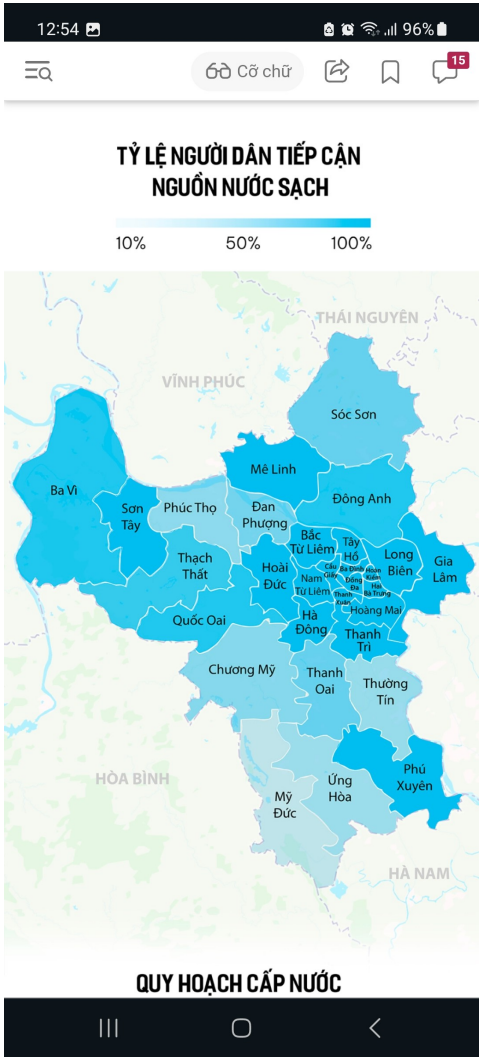


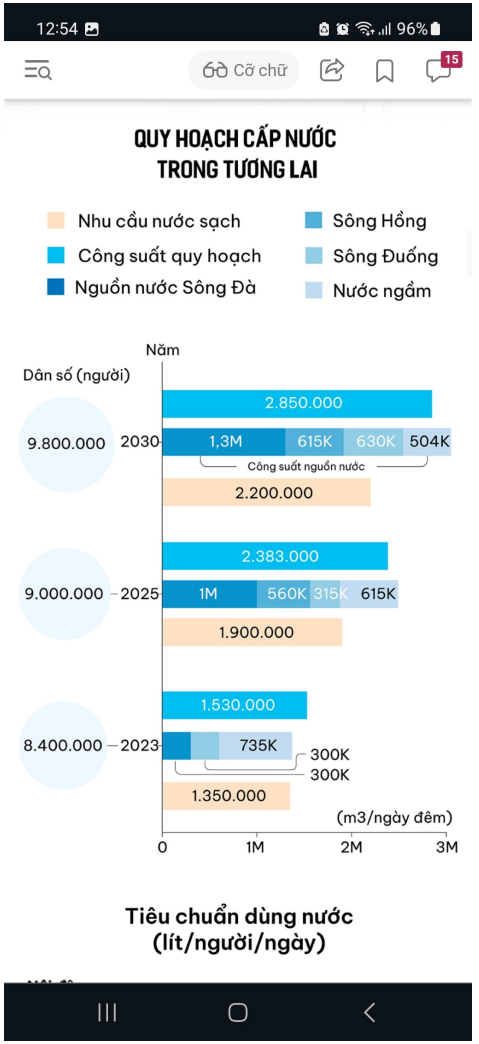


https://powerofbi.org/bad-and-good-charts-in-power-bi/

FOR EXAMPLE







https://vnexpress.net/chenh-lech-cung-cau-nuoc-sach-o-ha-noi-4669638.html

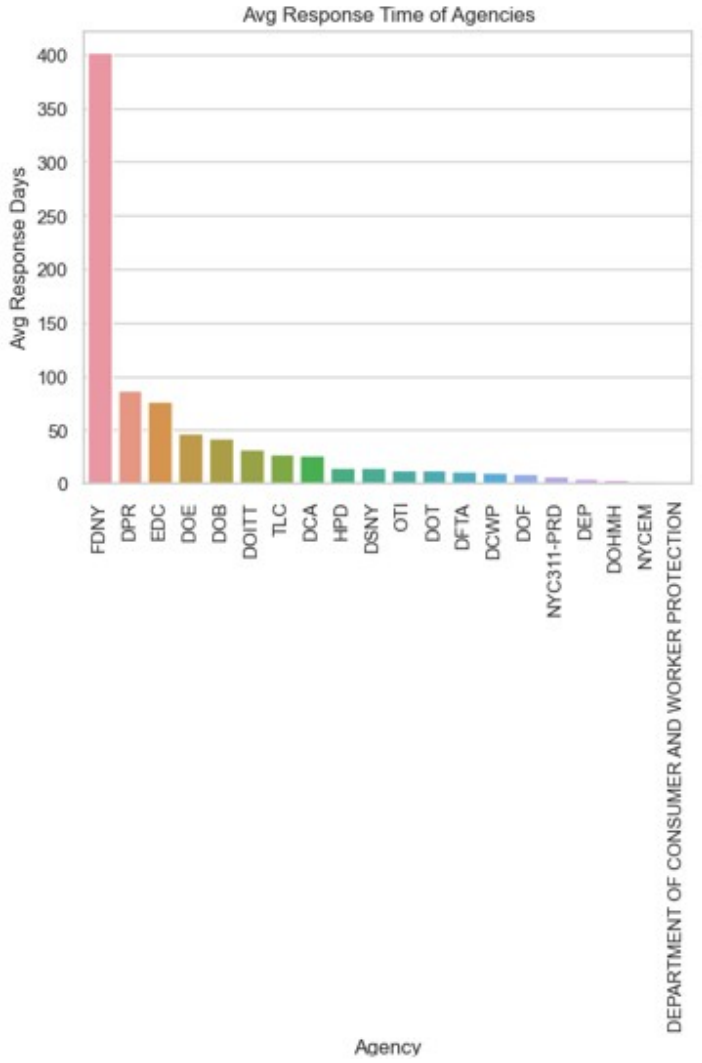
FOR EXAMPLE

Results: Attribute Analysis

Response Time

Agency with slowest response time:

- FDNY: Fire Department of NYC
- DPR: NYC Department of Park and Recreation
- EDC: NYC Economic Development Corporation

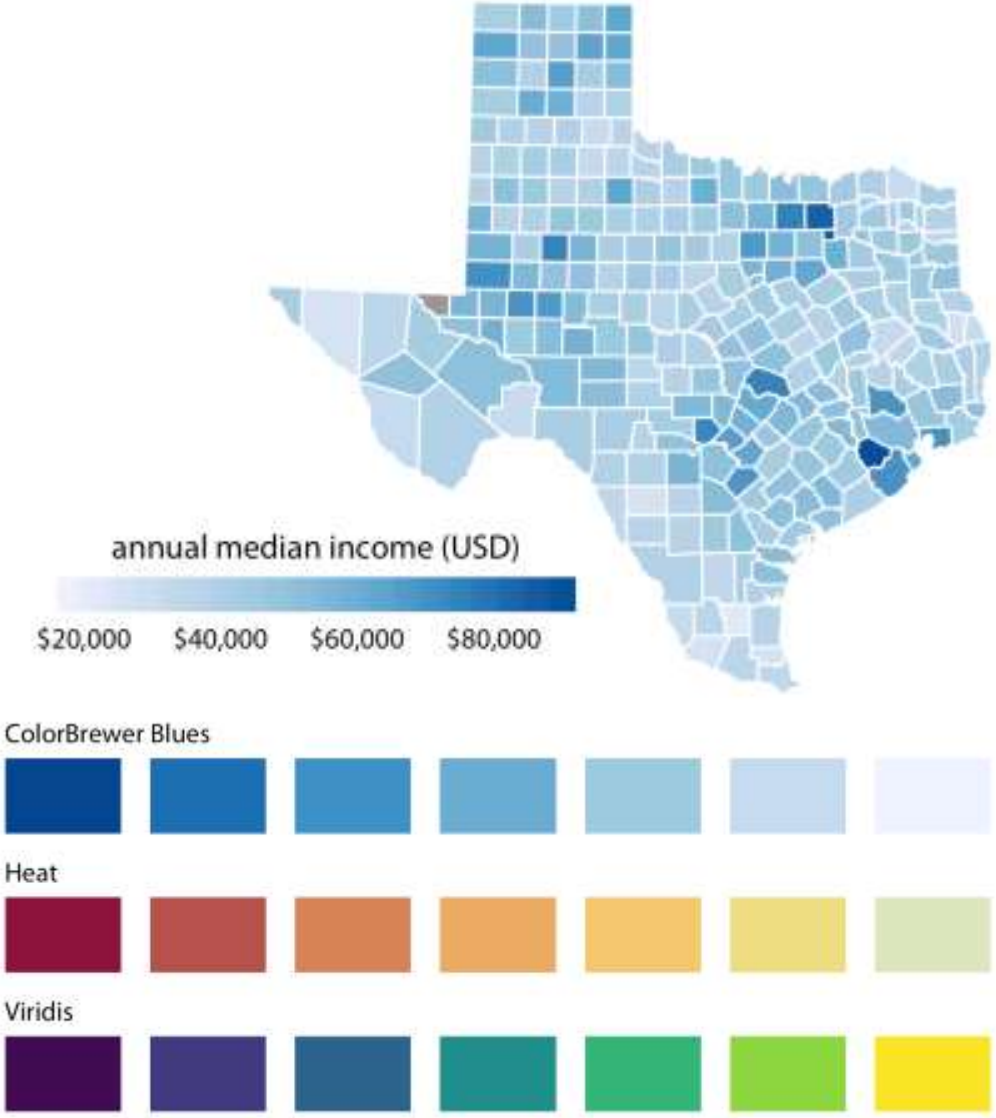


COLOR SCALE

- To represent data values
- To distinguish groups
- To highlight

COLOR TO REPRESENT VALUES

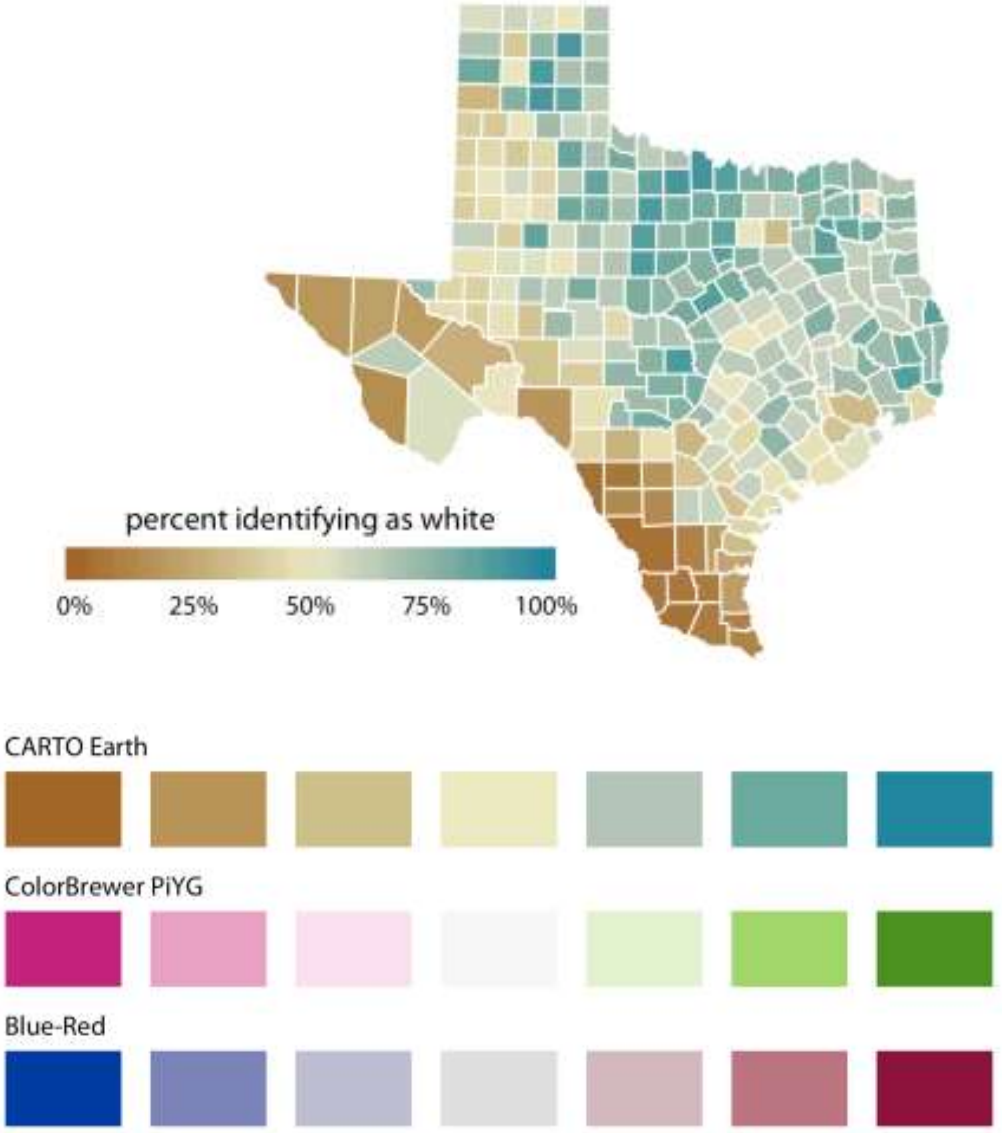
- Quantitative data values
  - E.g., income, temperature, speed
- Sequential color scale
  - Clearly indicates which values are larger or smaller than which other ones.
  - How distant two specific values are from each other.
  - The color scale needs to be perceived to vary uniformly across its entire range.



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

COLOR TO REPRESENT VALUES

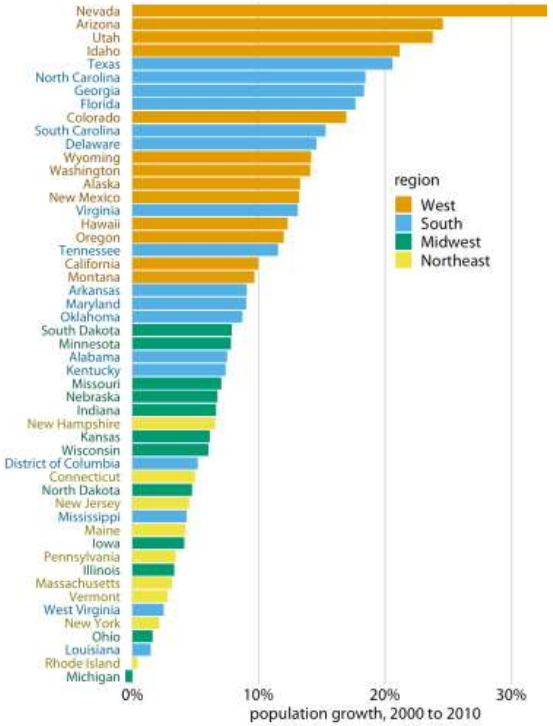
- Quantitative data values
  - E.g., income, temperature, speed
- Diverging color scale
  - One of two directions relative to a neutral midpoint.
  - How far in either direction it deviates from the midpoint.
  - The midpoint normally is a light color

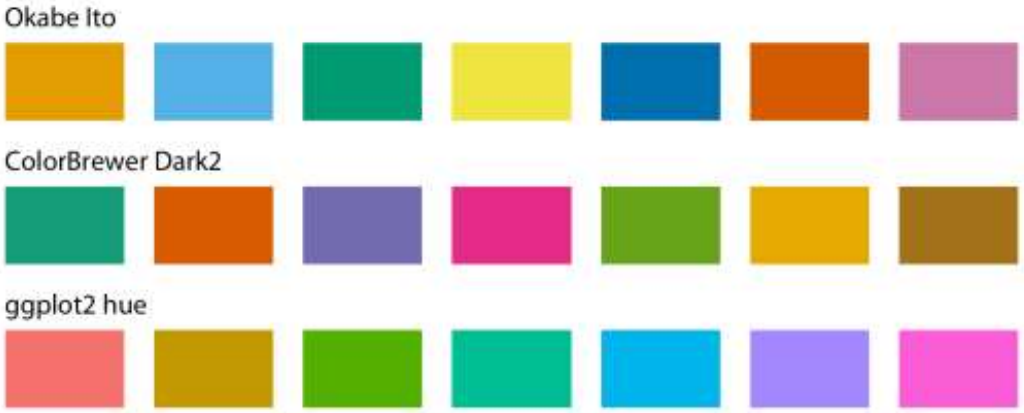


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COLOR TO DISTINGUISH GROUPS

- Groups without order
  - E.g., countries on a map, manufacturers of products
- Qualitative color scale
  - Specific colors to look clearly distinct from each other while also being equivalent to each other.
  - No one color should stand out relative to the others.
  - The colors should not create the impression of an order (e.g., successively lighter)

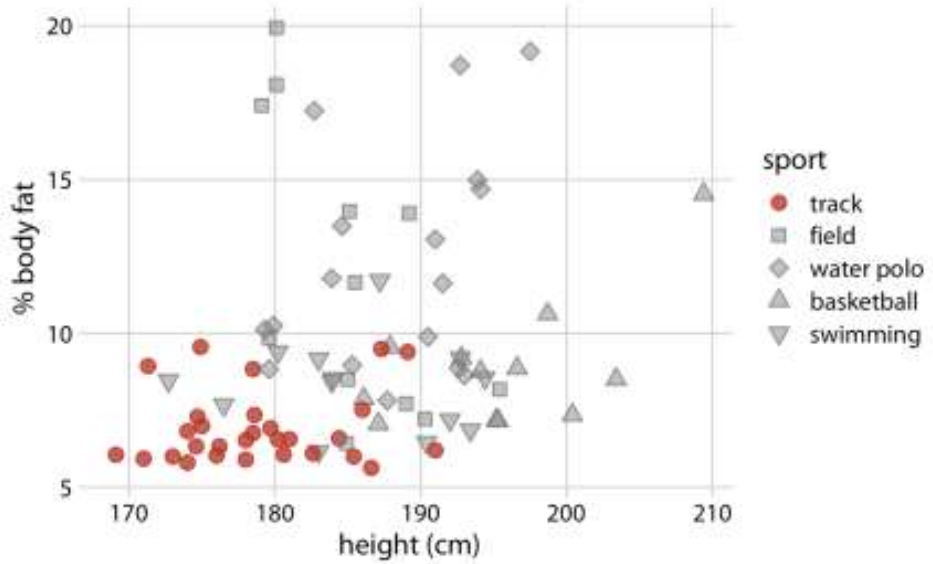


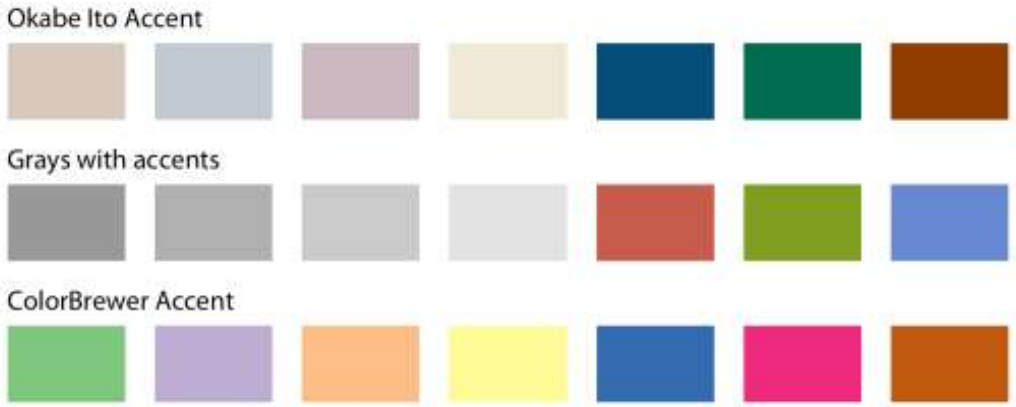


Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

COLOR TO HIGHLIGHT

- To highlight specific elements in the data.
- Accent color scale
  - Vividly stands out against the rest of the figure.
  - Contains both a set of subdued colors and a matching set of stronger, darker, and/or more saturated colors.
- It is critical that the baseline colors do not compete for attention.

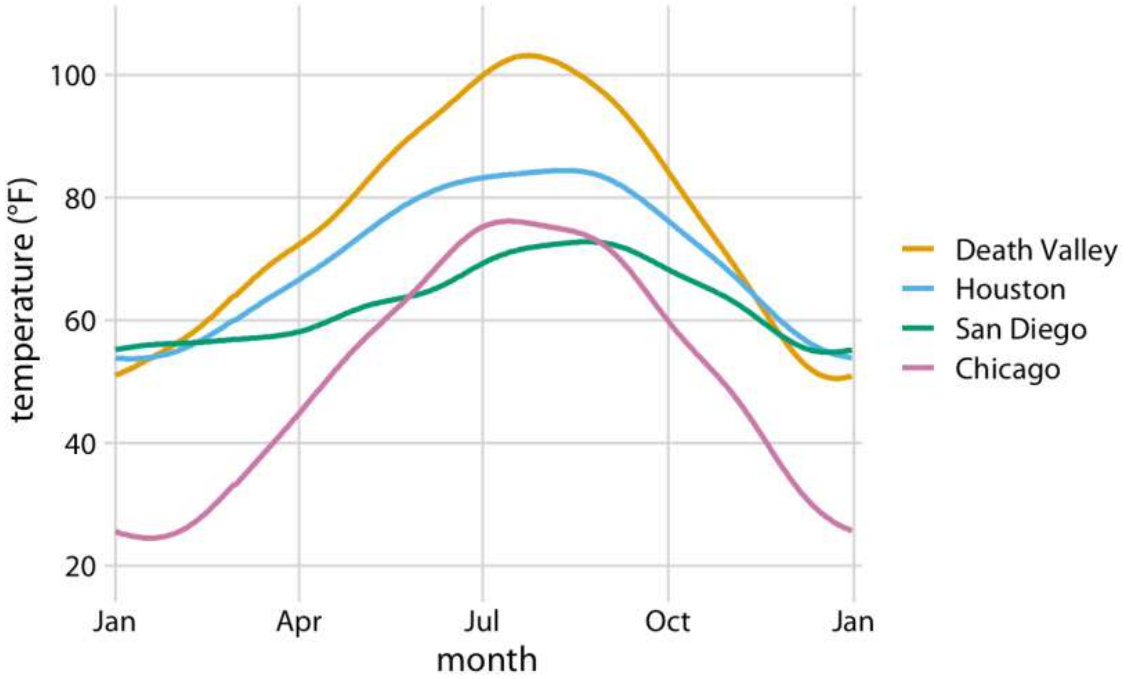


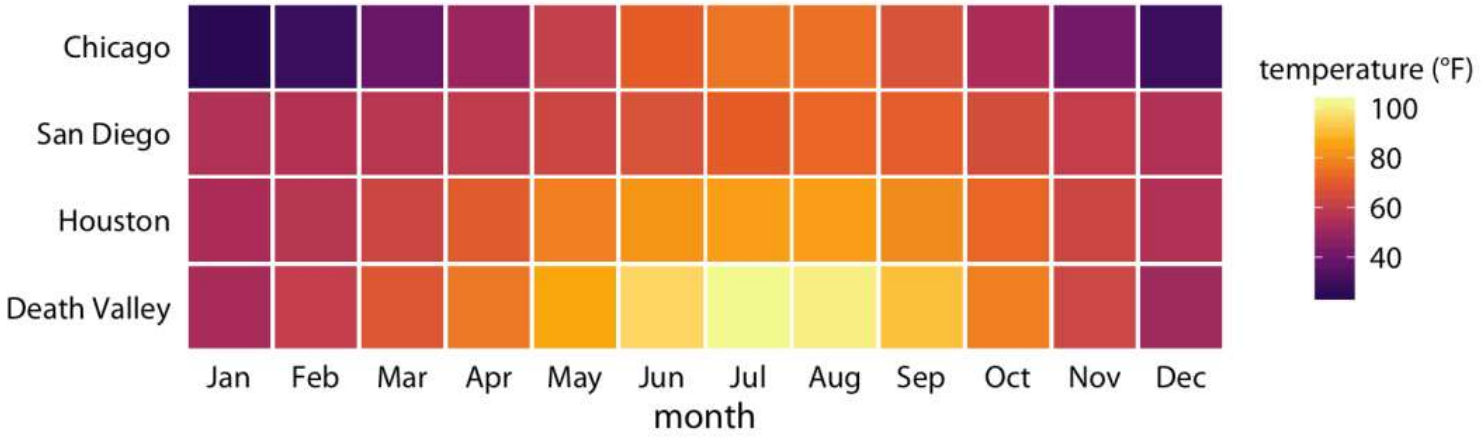


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FOR INSTANCE

Month	Day	Location	Station ID	Temperature (°F)
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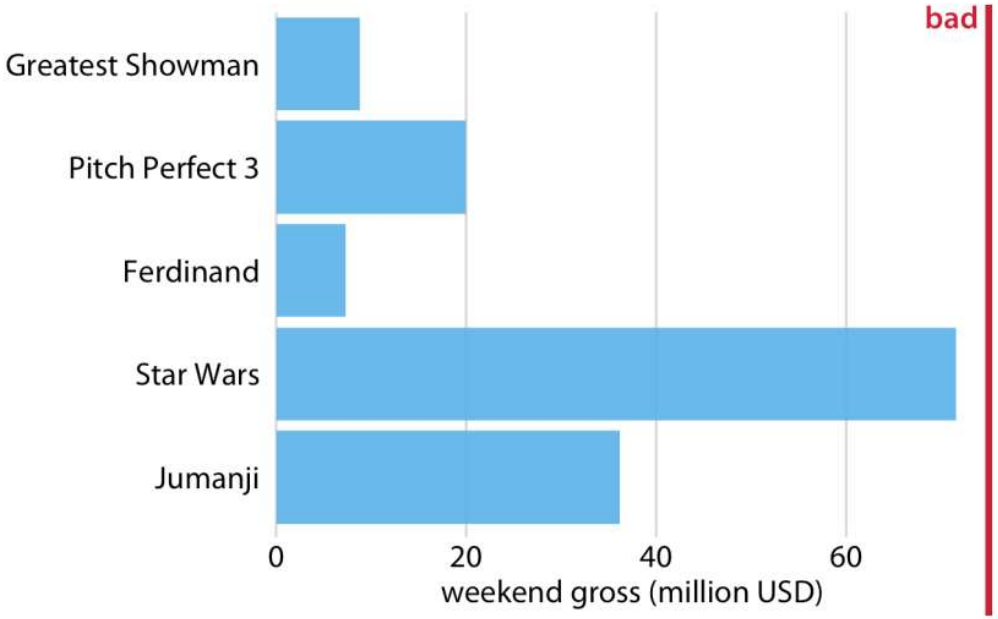
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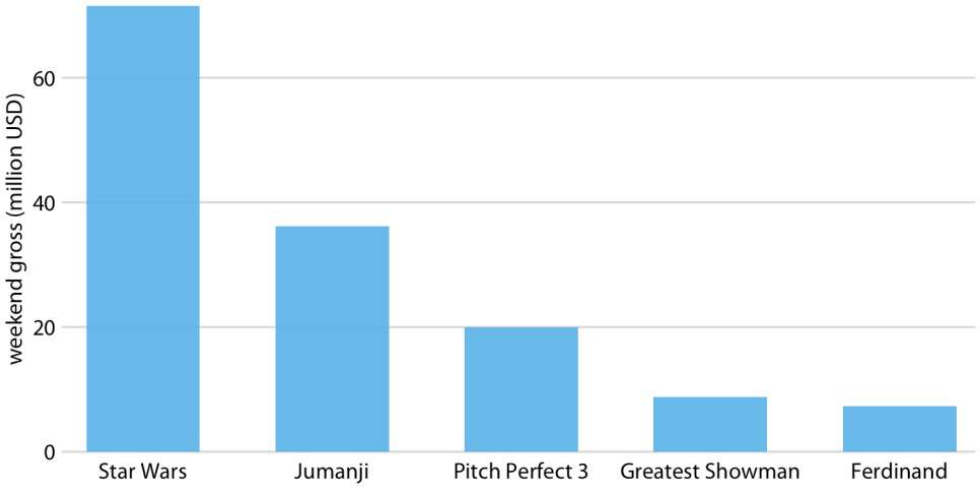
VISUALIZING AMOUNTS

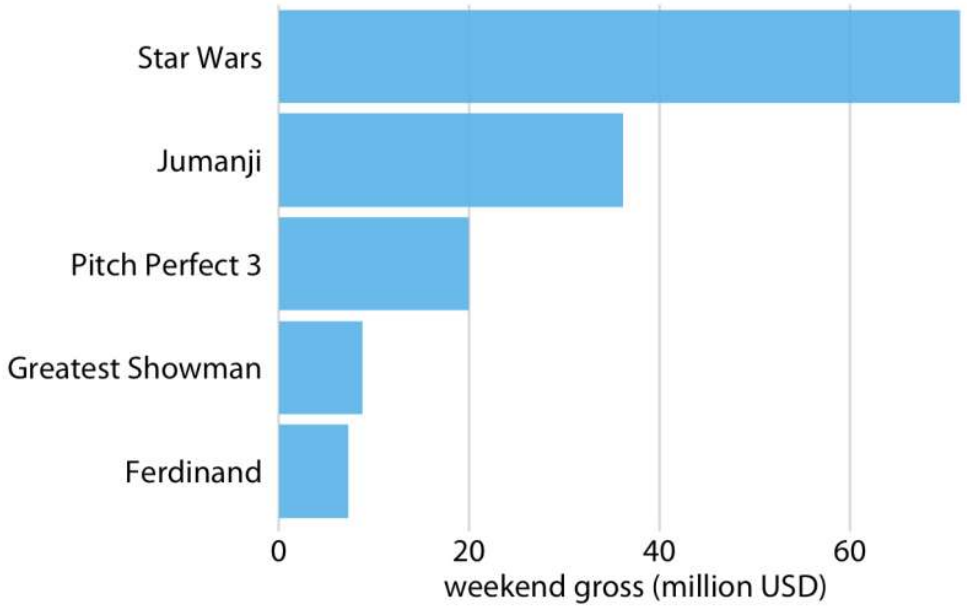
- When we are interested in the magnitude of some set of numbers.
  - E.g., total sales of different brands, total customer in different areas
- Standard visualization
  - Simple bars
  - Grouped bars
  - Stacked bars
- Alternatives
  - Dot plot
  - Heatmap

SIMPLE BARS

Rank	Title	Weekend gross
1	*Star Wars: The Last Jedi*	$71,565,498
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4	*The Greatest Showman*	$8,805,843
5	*Ferdinand*	$7,316,746

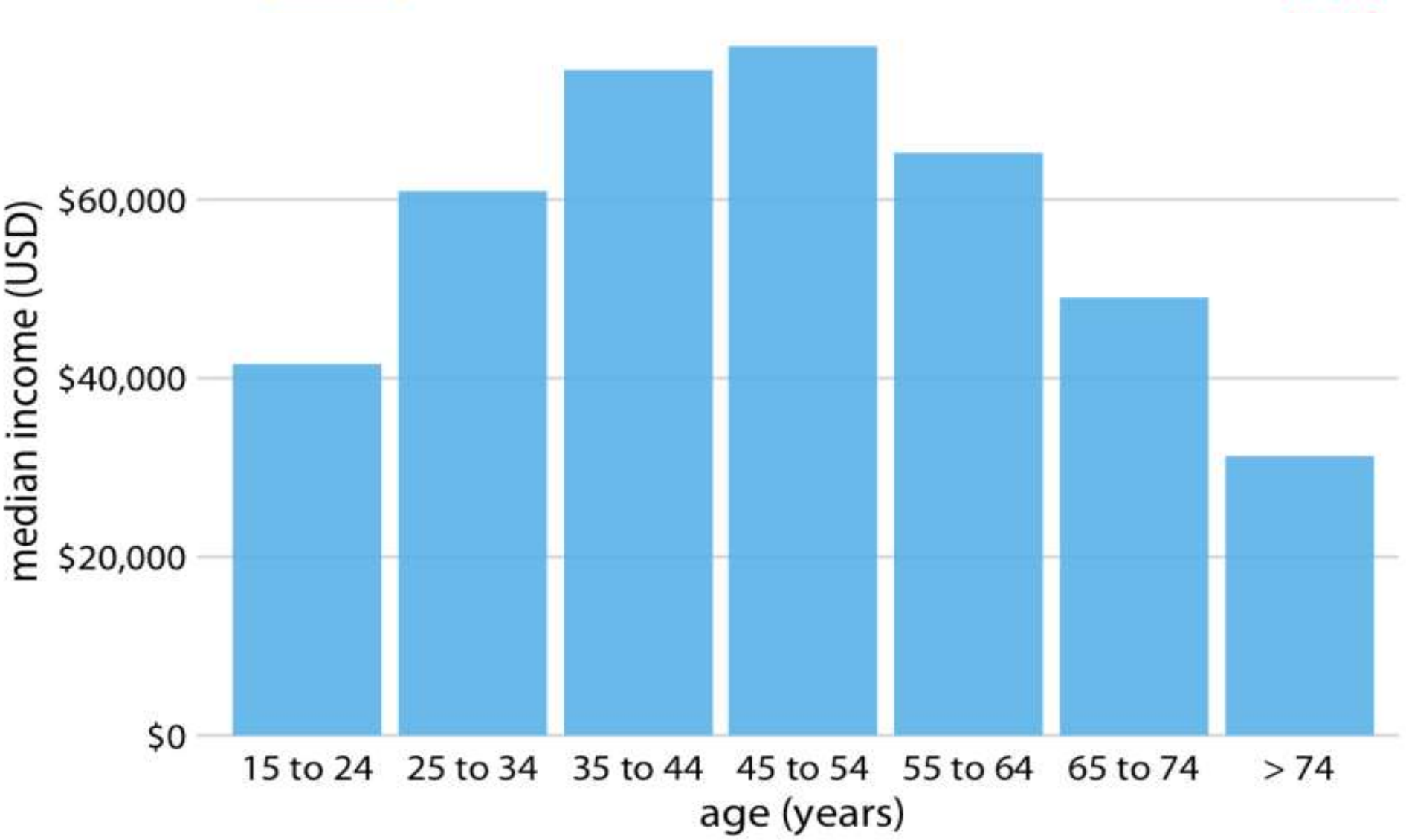






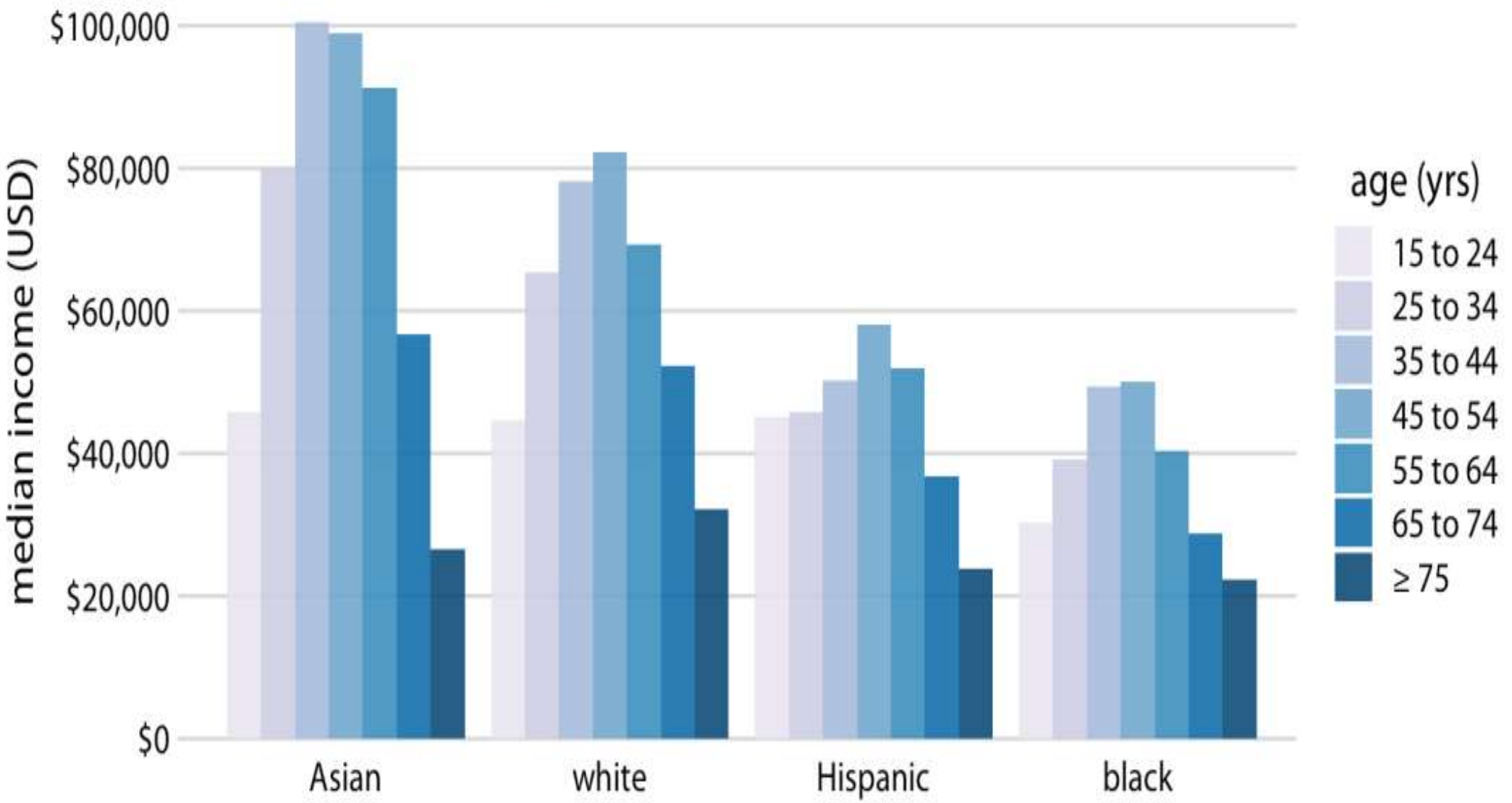
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SIMPLE BARS



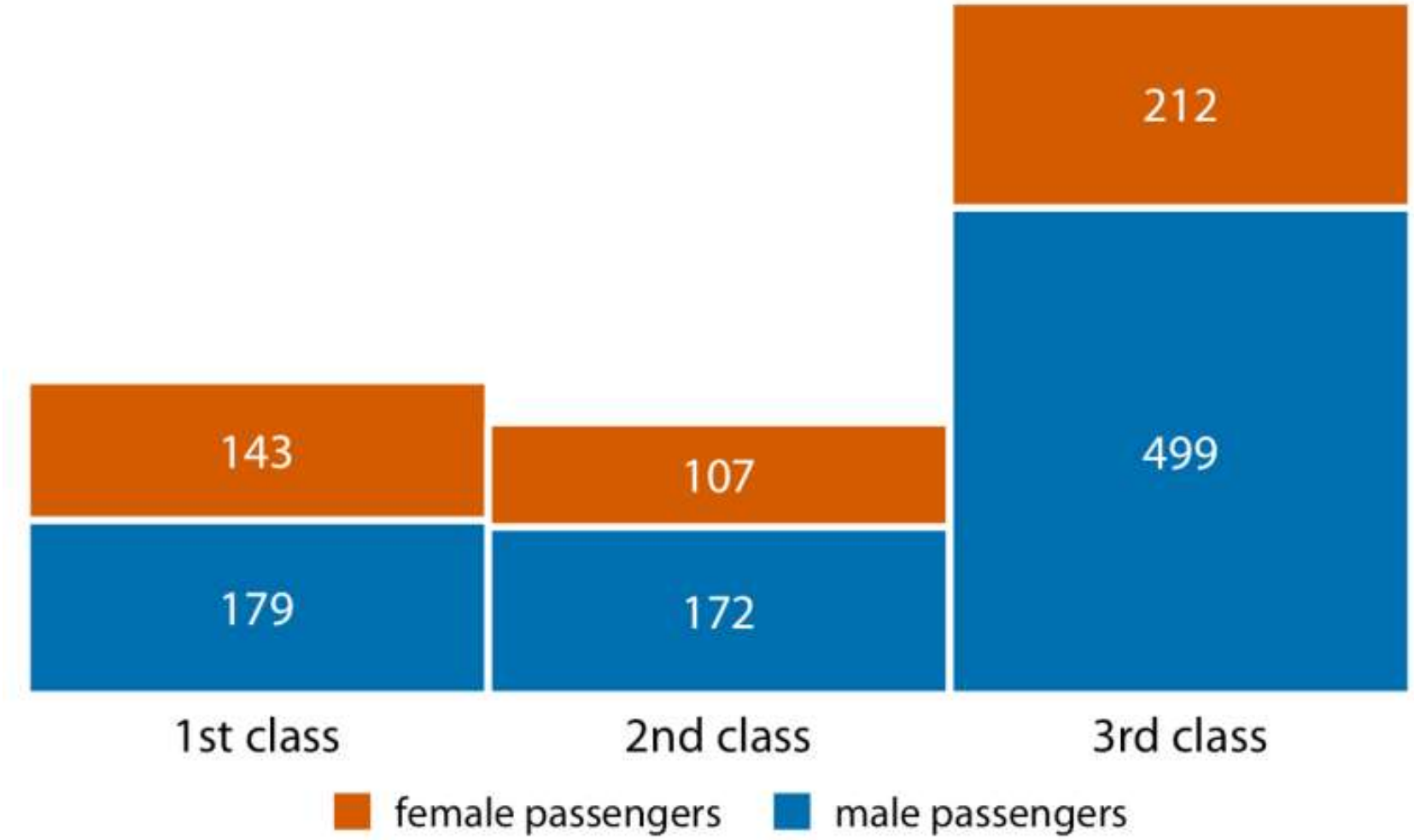
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

GROUP BARS



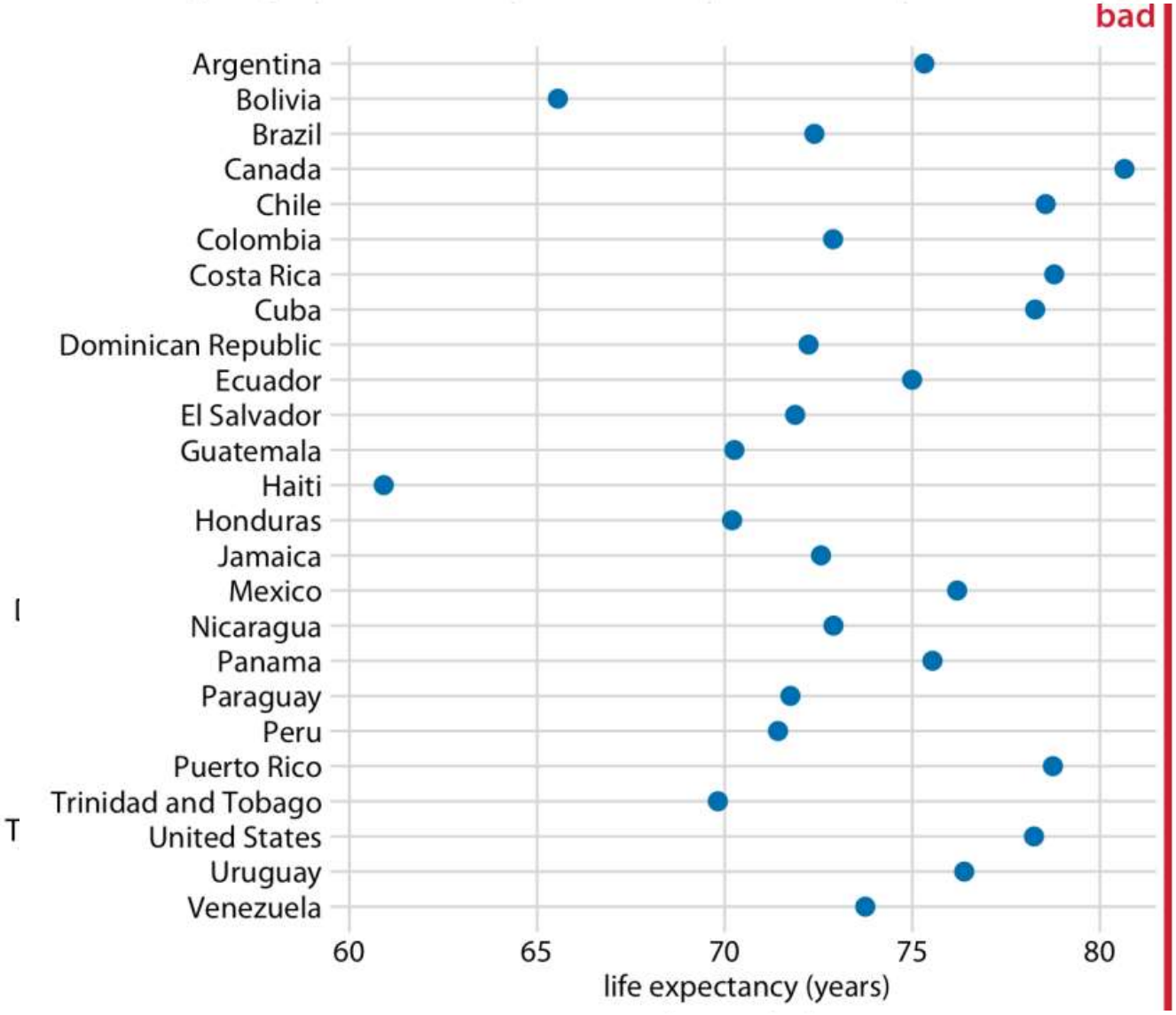
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

STACKED BARS



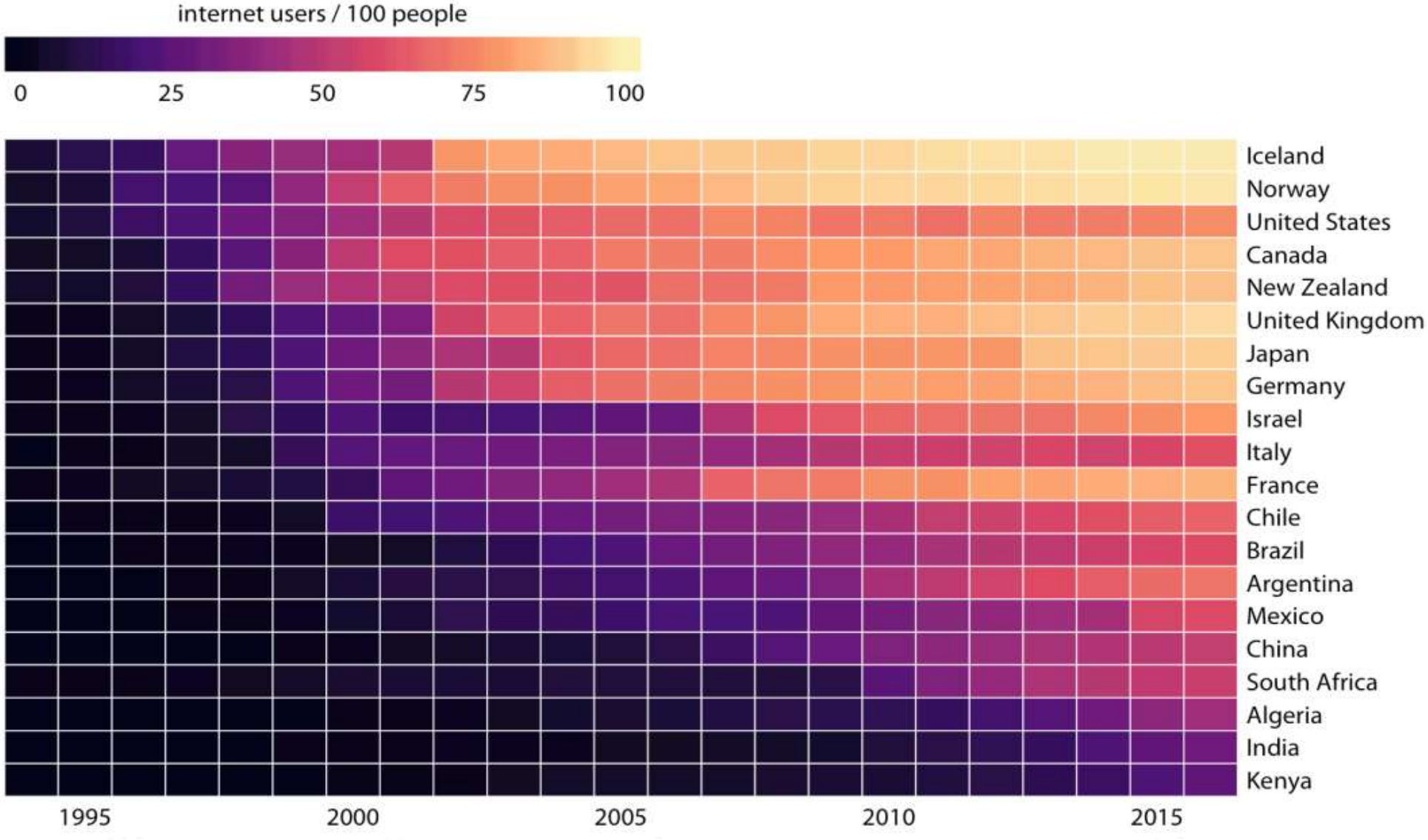
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

DOT PLOTS



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

DOT PLOTS

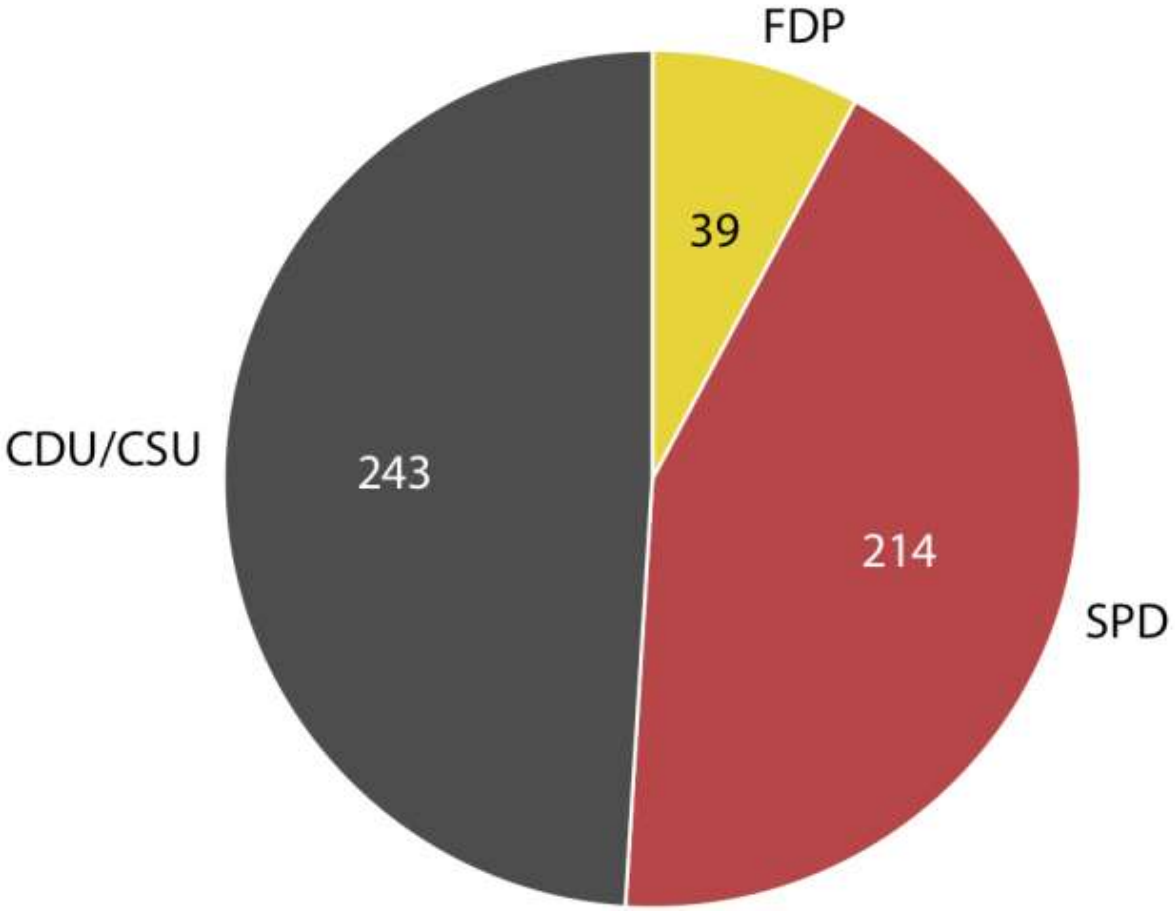


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VISUALIZING PROPORTIONS

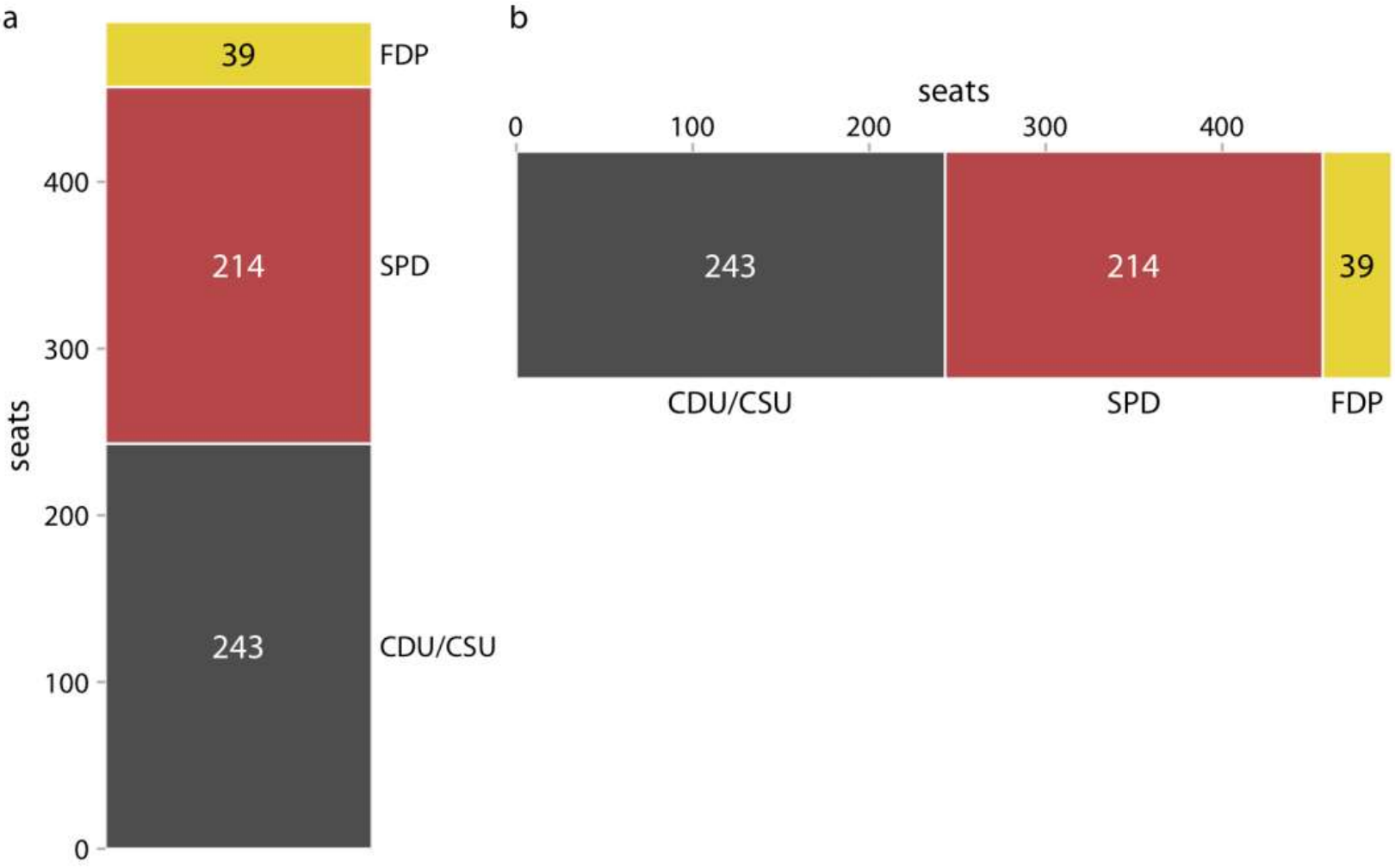
- How some group, entity, or amount breaks down into individual pieces that each represent a proportion of the whole.
  - Pie charts
  - Stacked bars
  - Stacked densities

PIE CHARTS



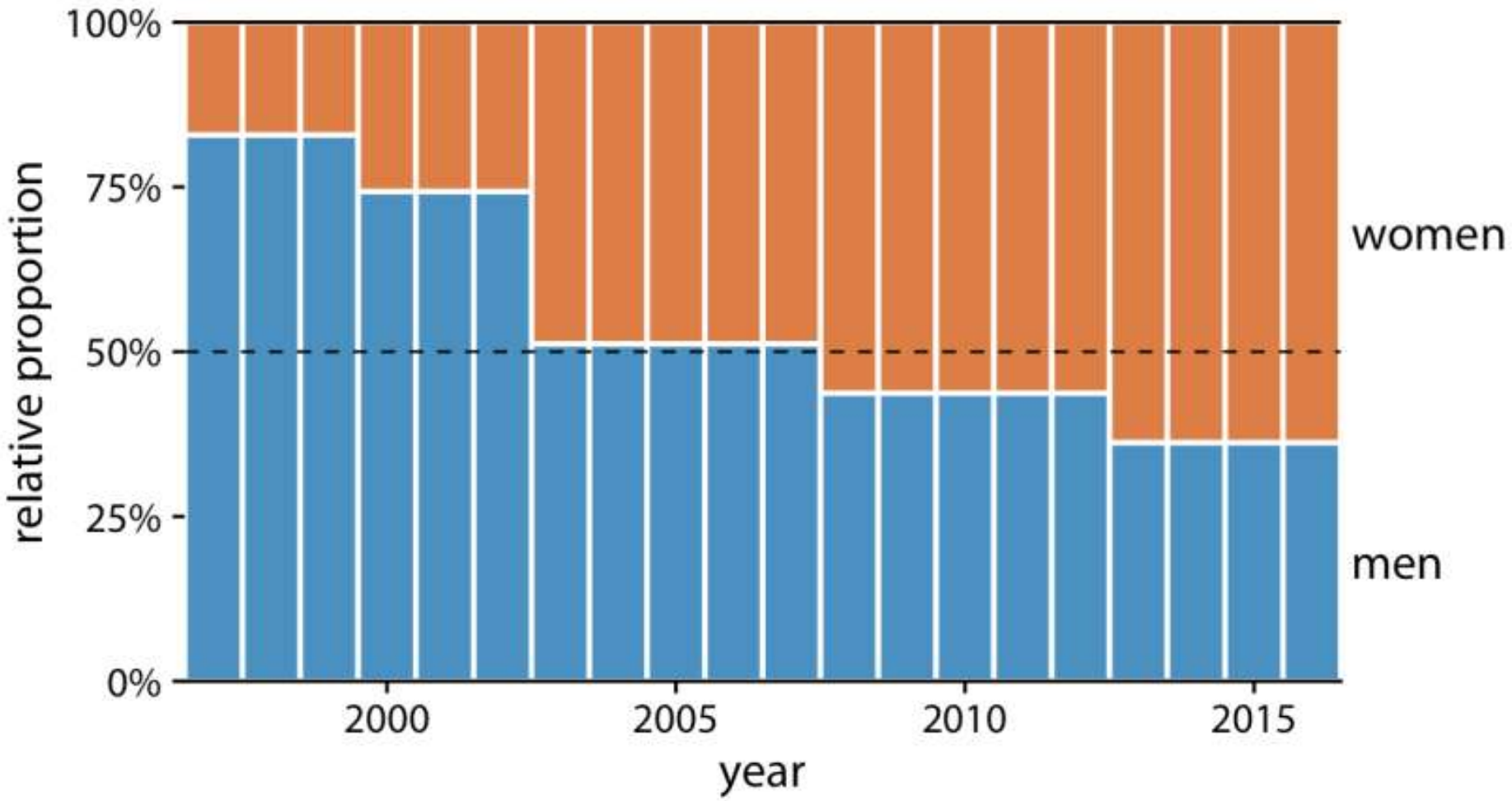
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

STACKED BARS



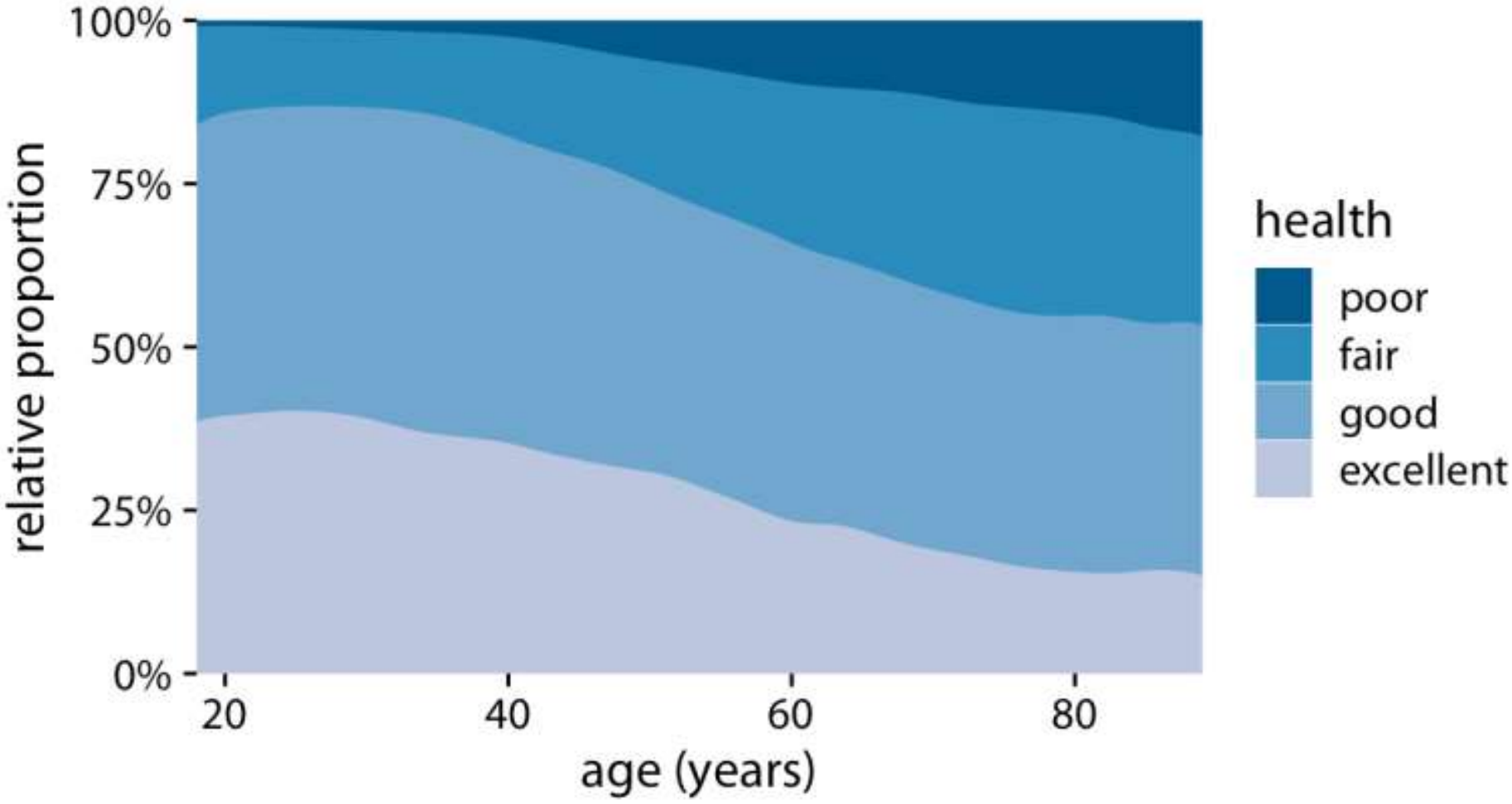
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

STACKED BARS



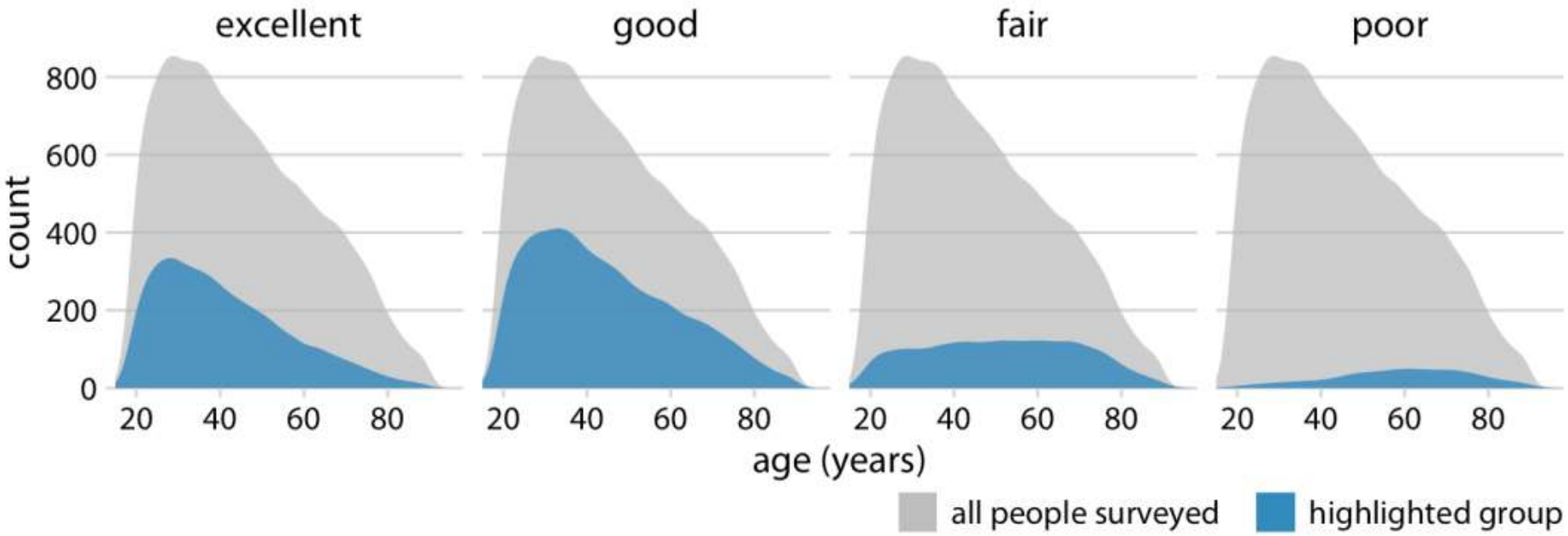
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

STACKED DENSITIES



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

STACKED DENSITIES

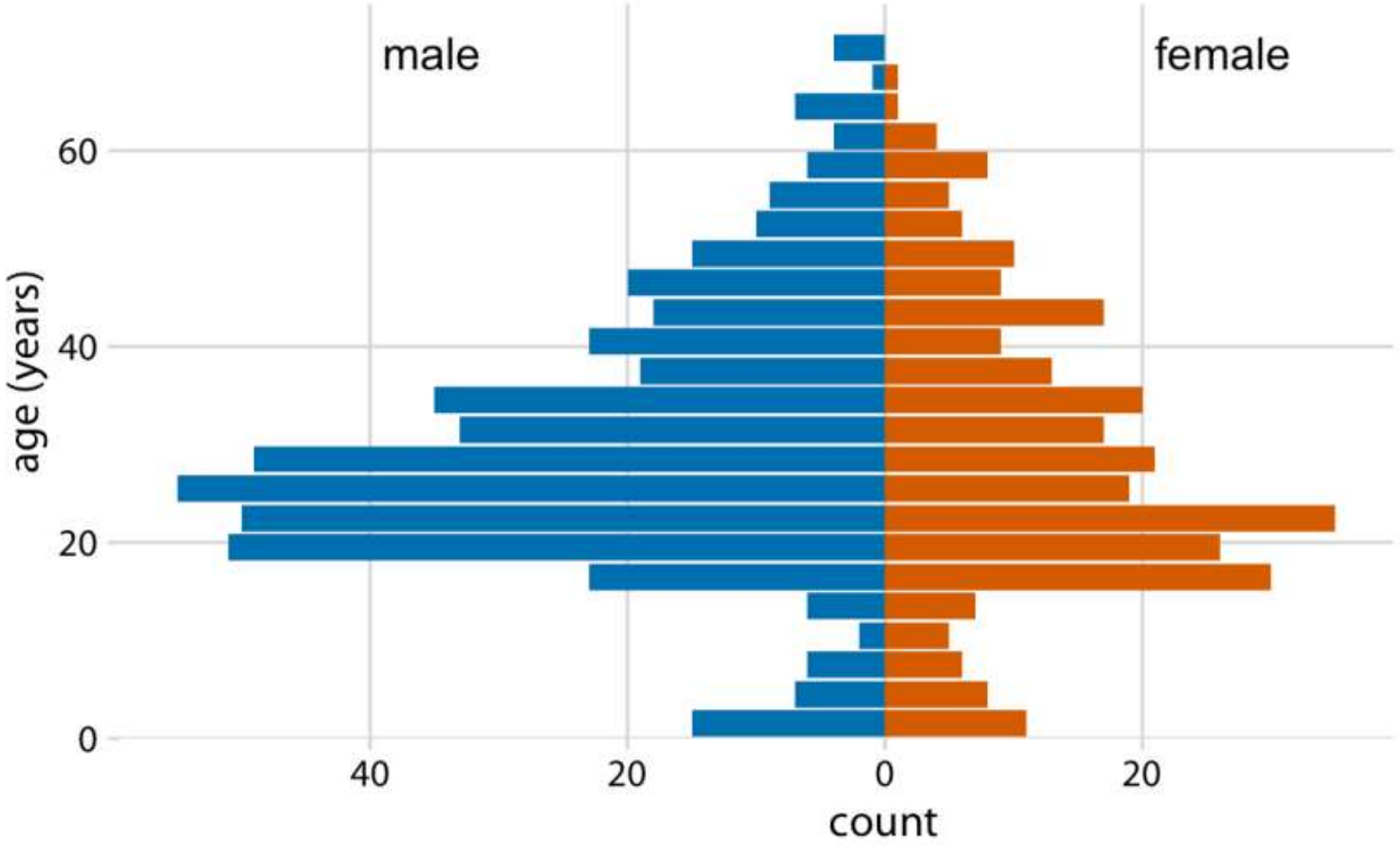


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VISUALIZING DISTRIBUTIONS

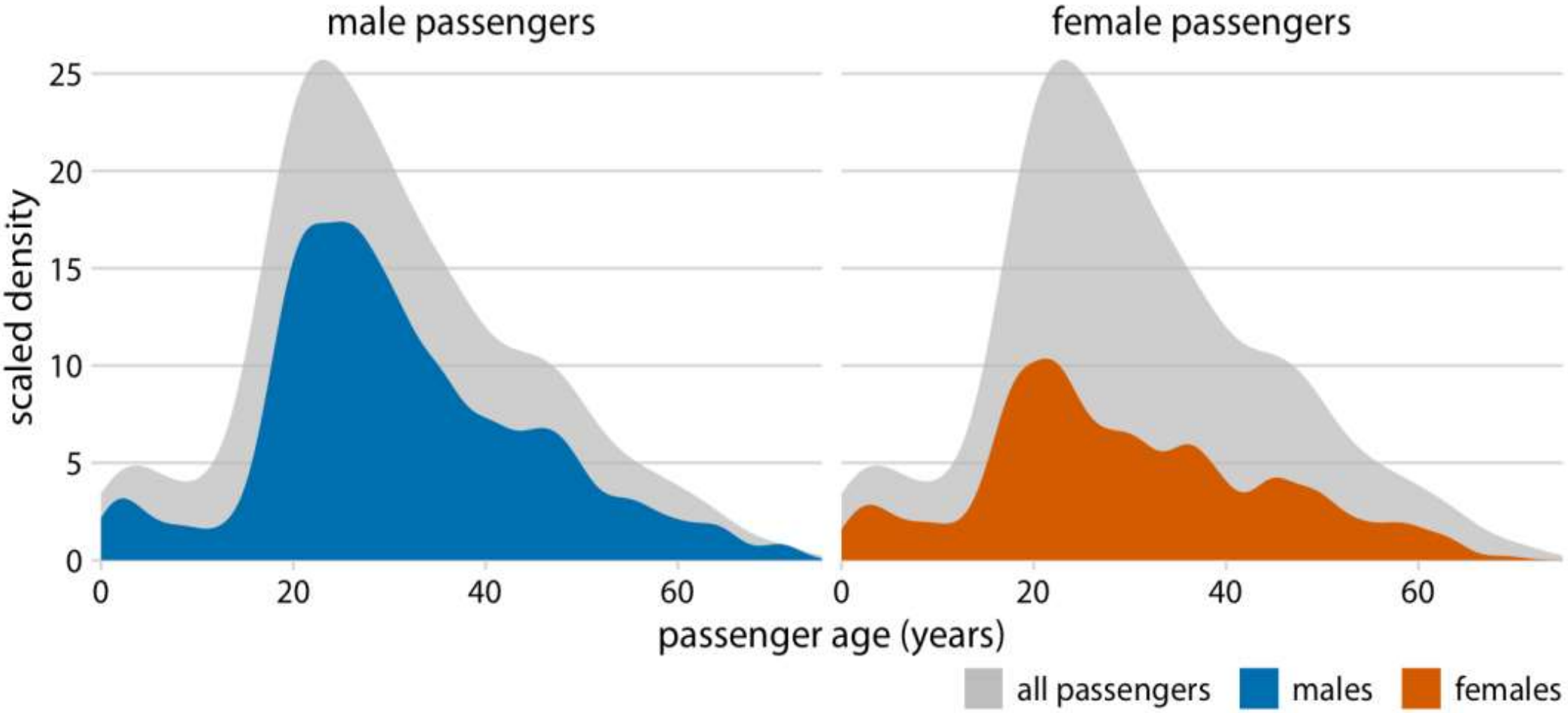
- To understand how a particular variable is distributed in a dataset.
  - Histograms
  - Density plots

HISTOGRAMS



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DENSITY PLOTS

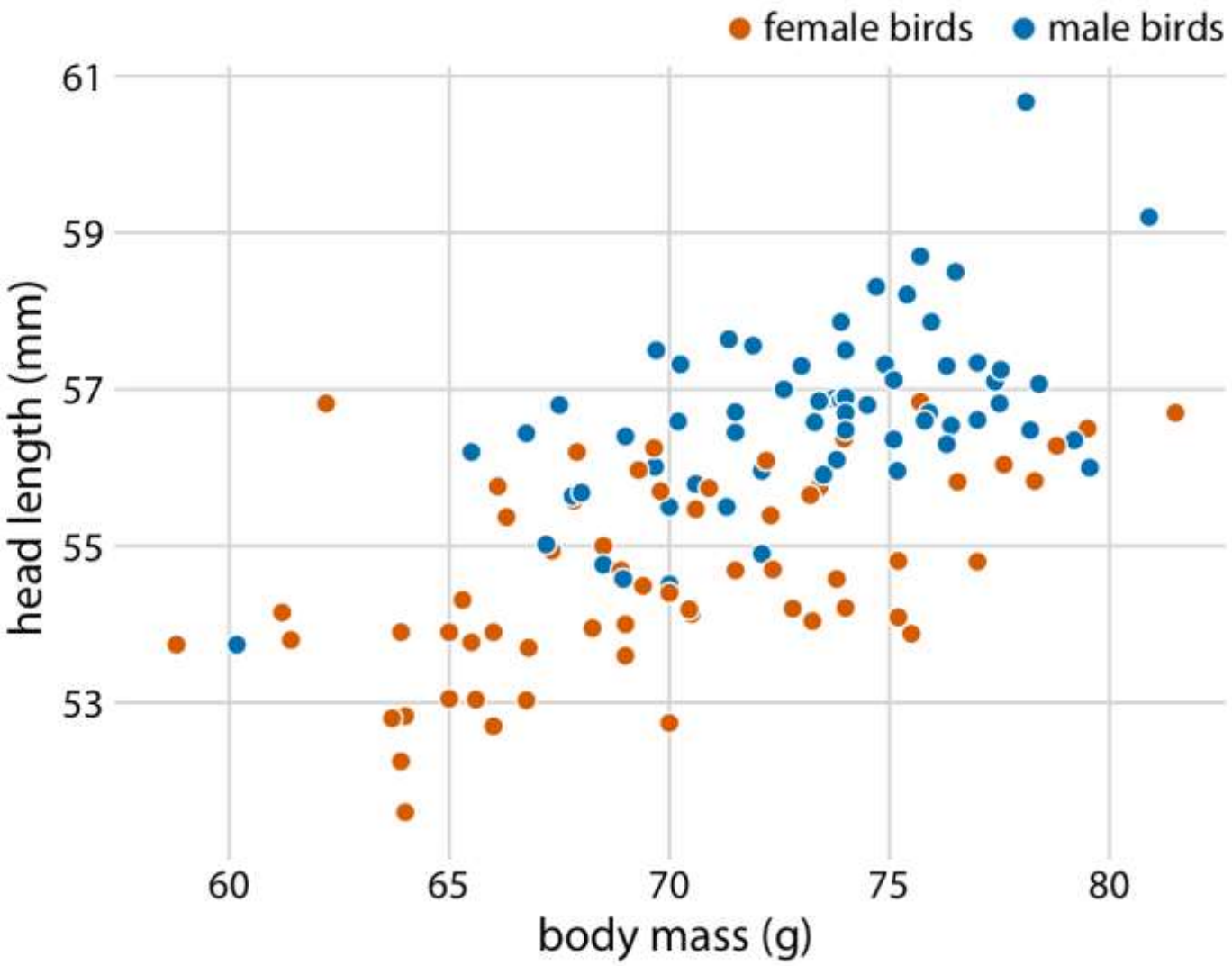


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VISUALIZING ASSOCIATIONS

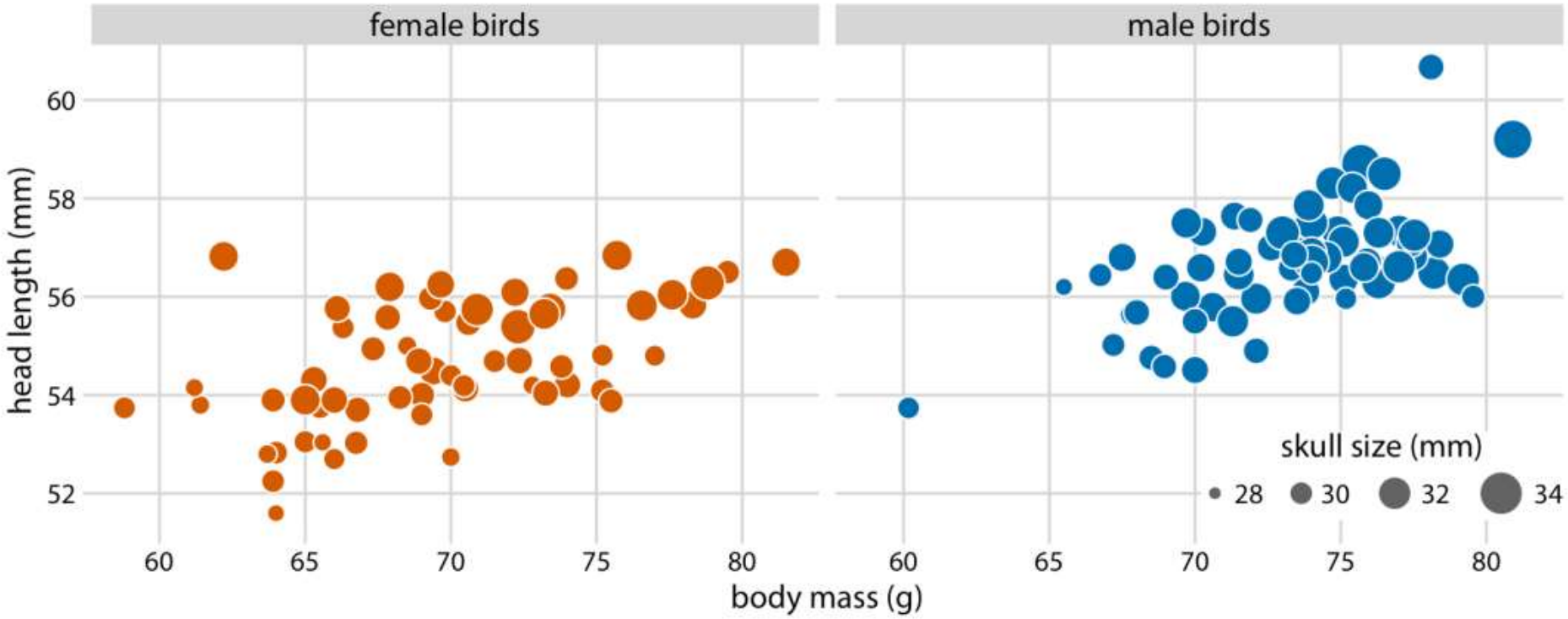
- How these variables relate to each other
  - E.g., height and weight
- Report
  - Scatterplot
  - Bubble chart
  - Scatterplot matrix
  - Correlogram

SCATTERPLOT



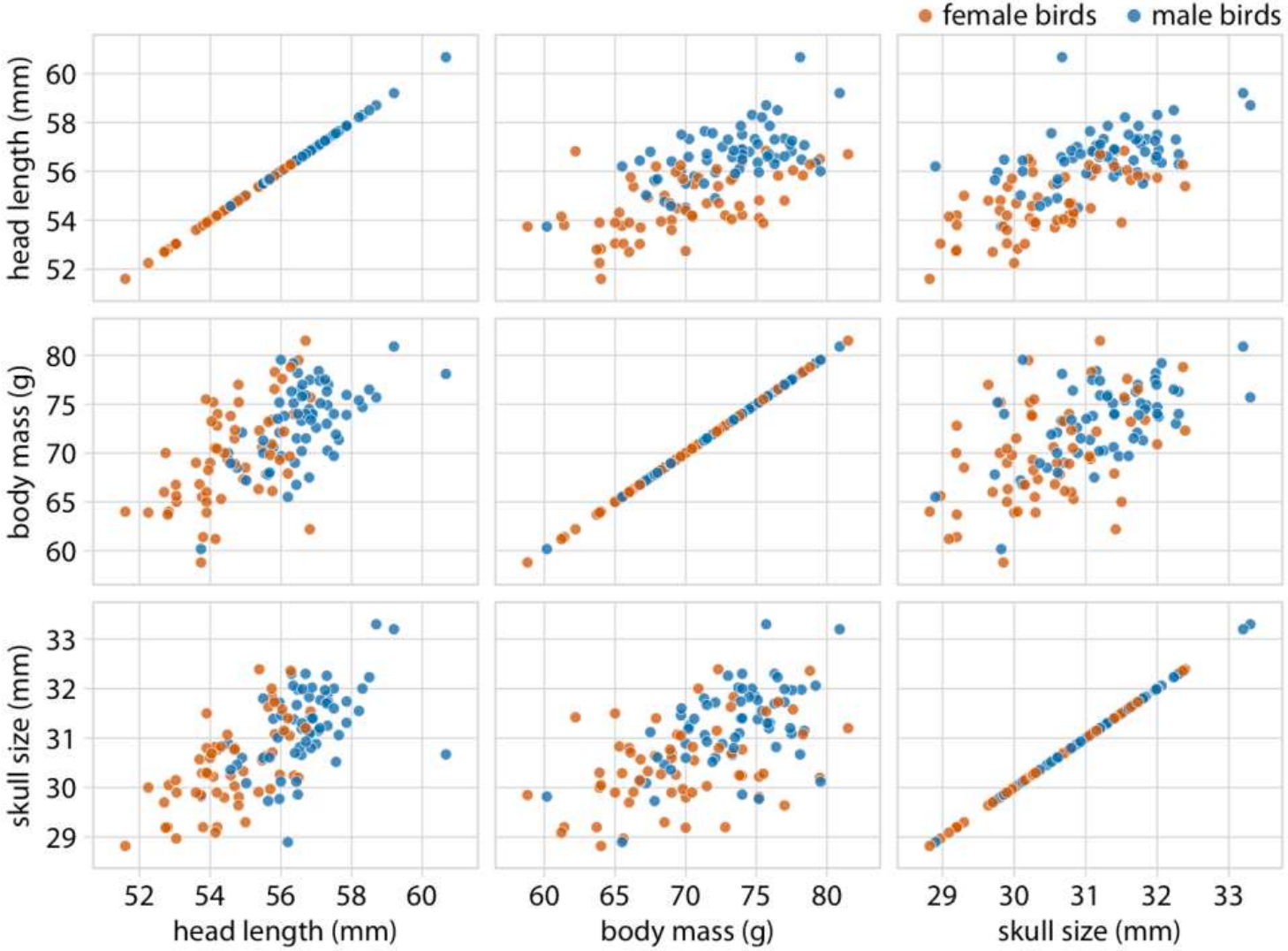
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BUBBLE CHART



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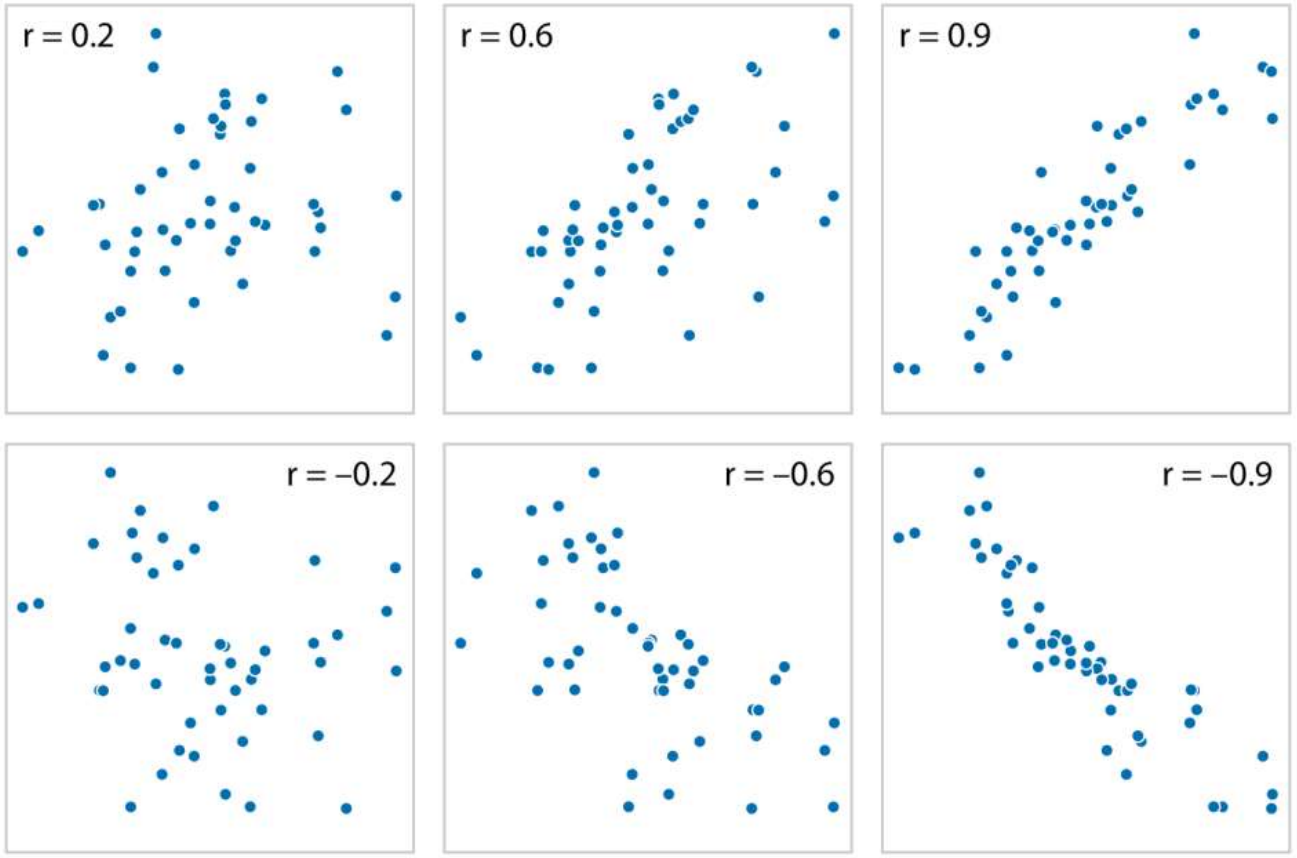
SCATTERPLOT MATRIX



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

CORRELOGRAMS

$$r = \frac{\Sigma_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma_i (x_i - \bar{x})^2} \sqrt{\Sigma_i (y_i - \bar{y})^2}}$$

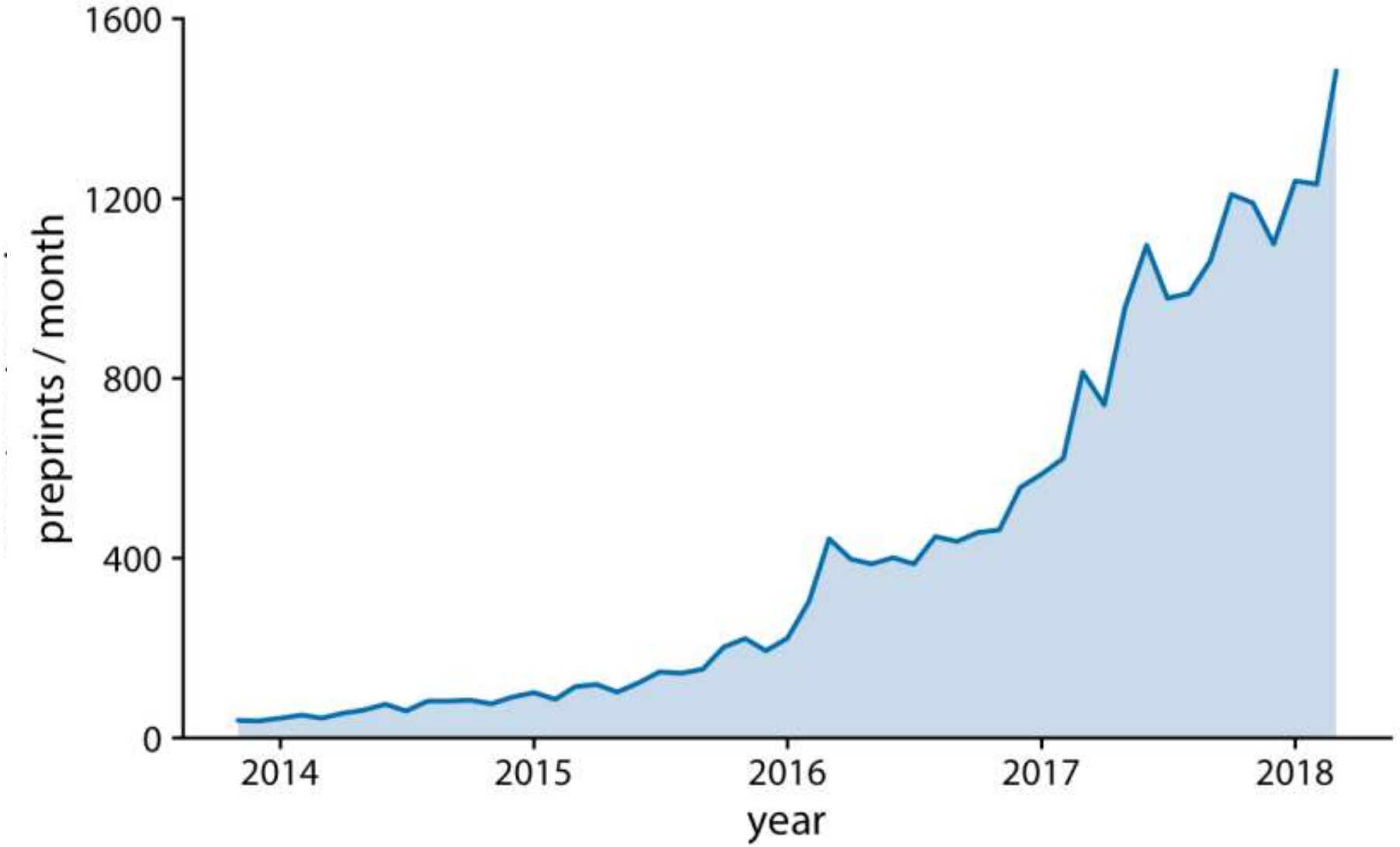


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VISUALIZING TIME SERIES

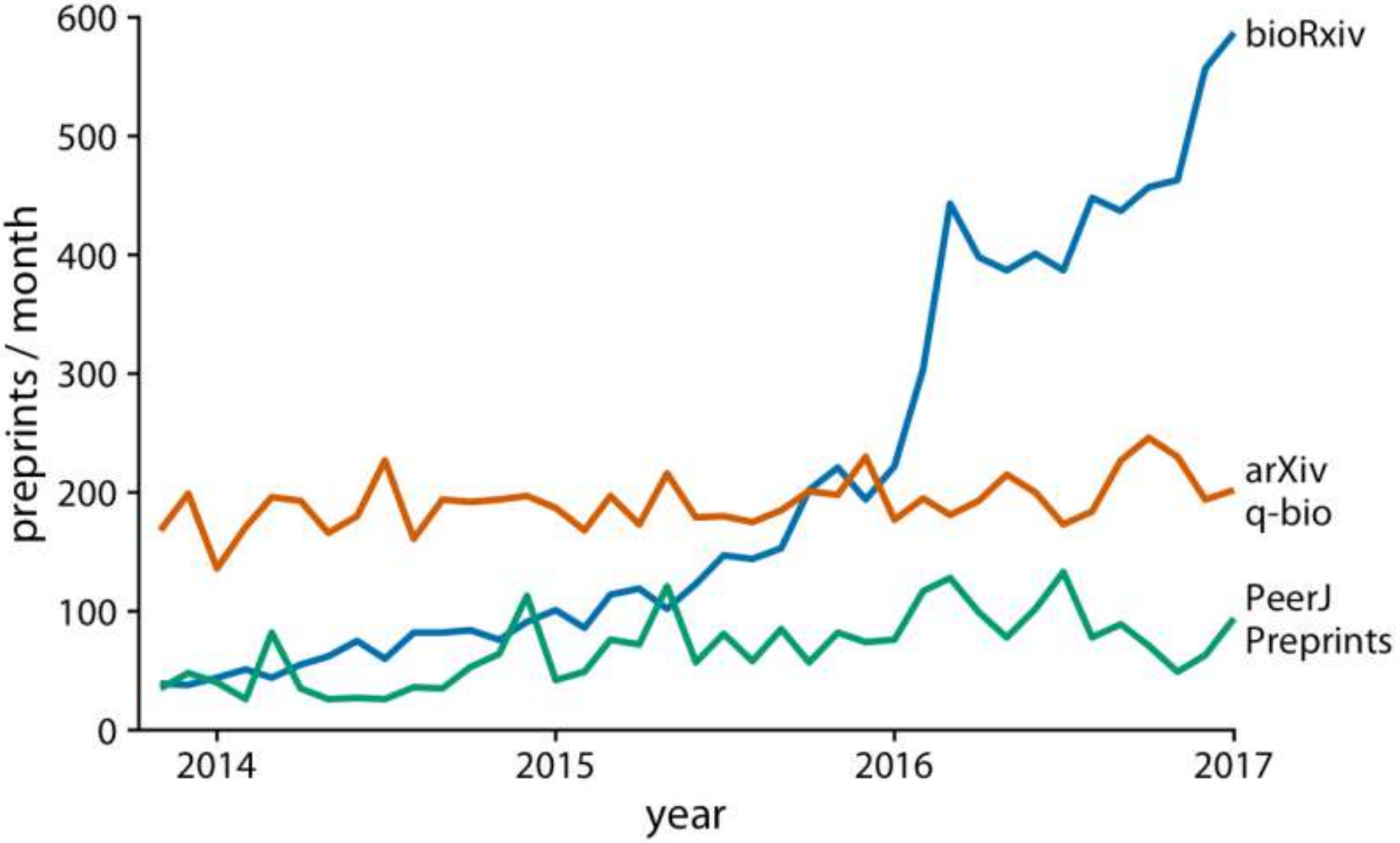
- To see pattern by time
  - Single time series
  - Multiple time series
  - Time series with two or more variables

SINGLE TIME SERIES



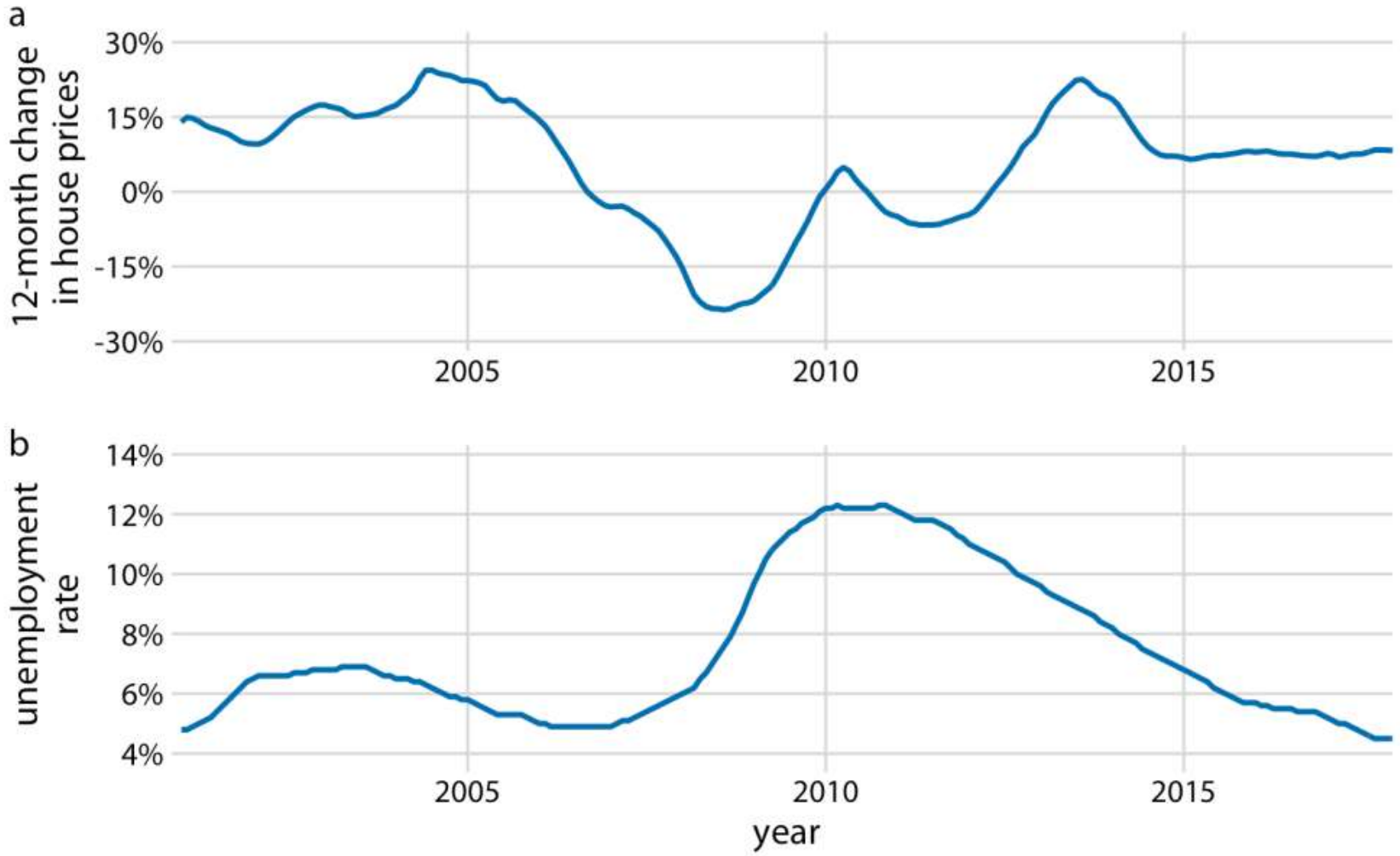
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MULTIPLE TIME SERIES



Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

TIME SERIES WITH TWO VARIABALES

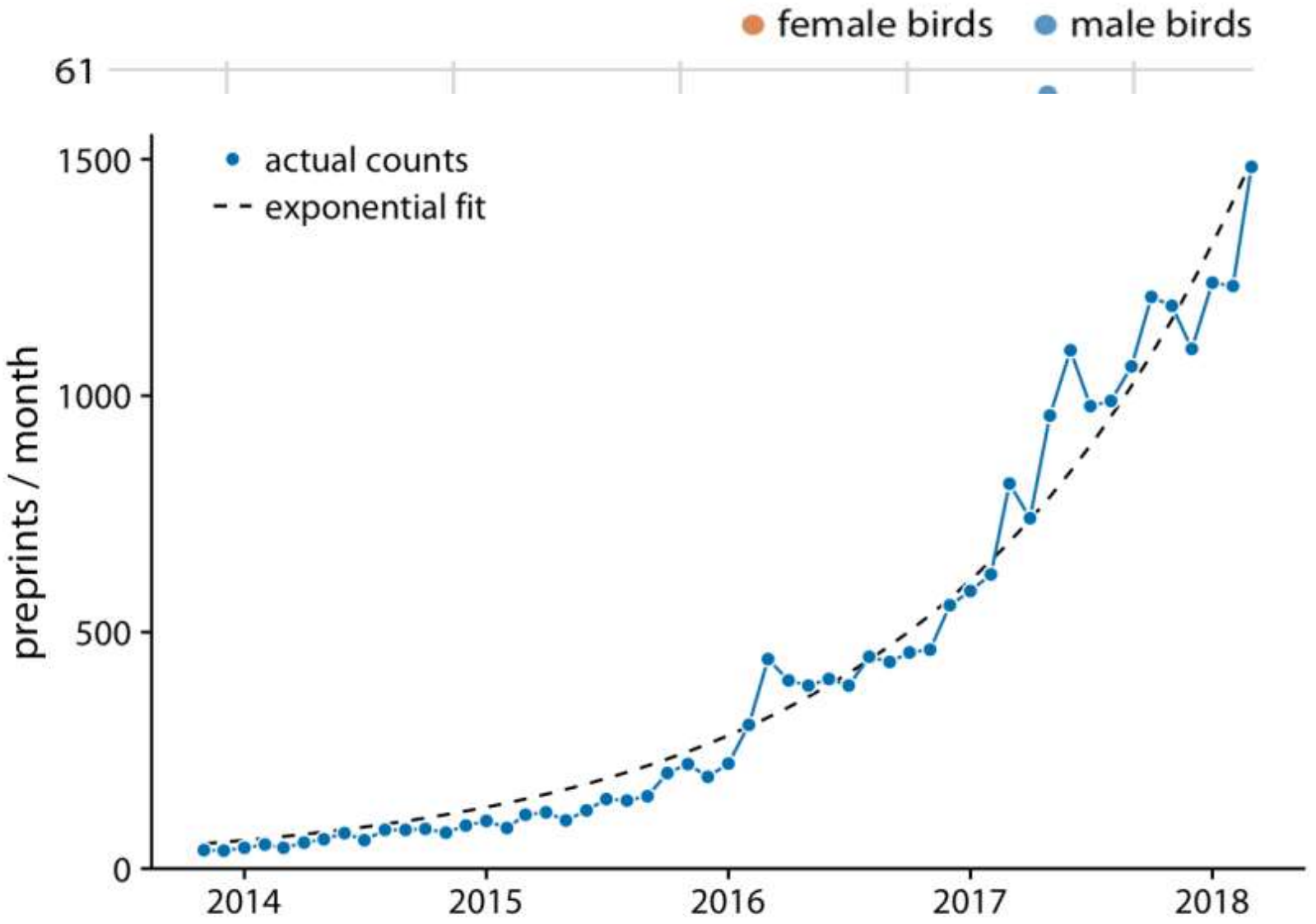


Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

VISUALIZING TRENDS

- To see key features of data
  - Smoothing
  - Linear
  - Curve

TIME SERIES WITH TWO VARIABALES

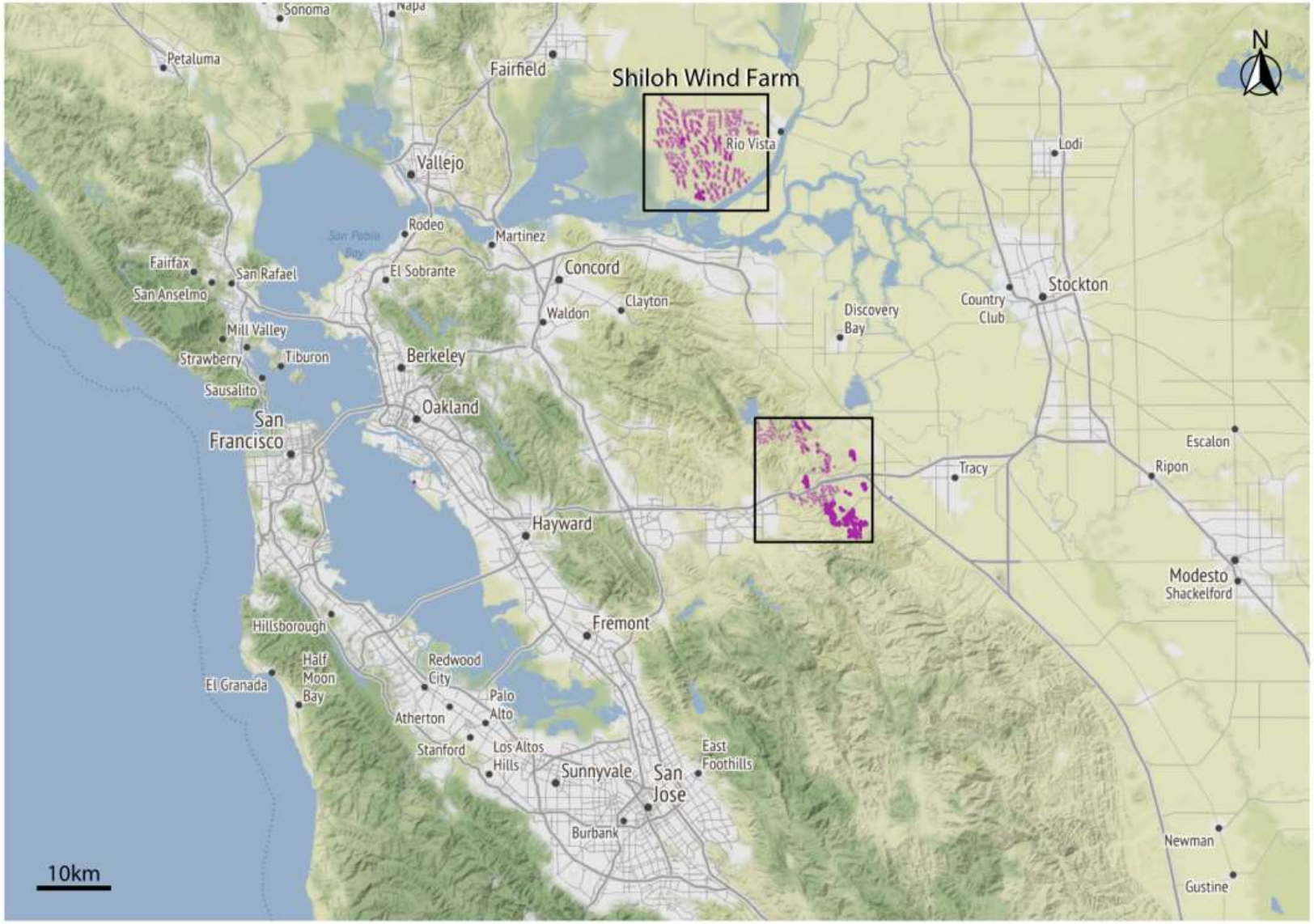


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VISUALIZING GEOSPATIAL DATA

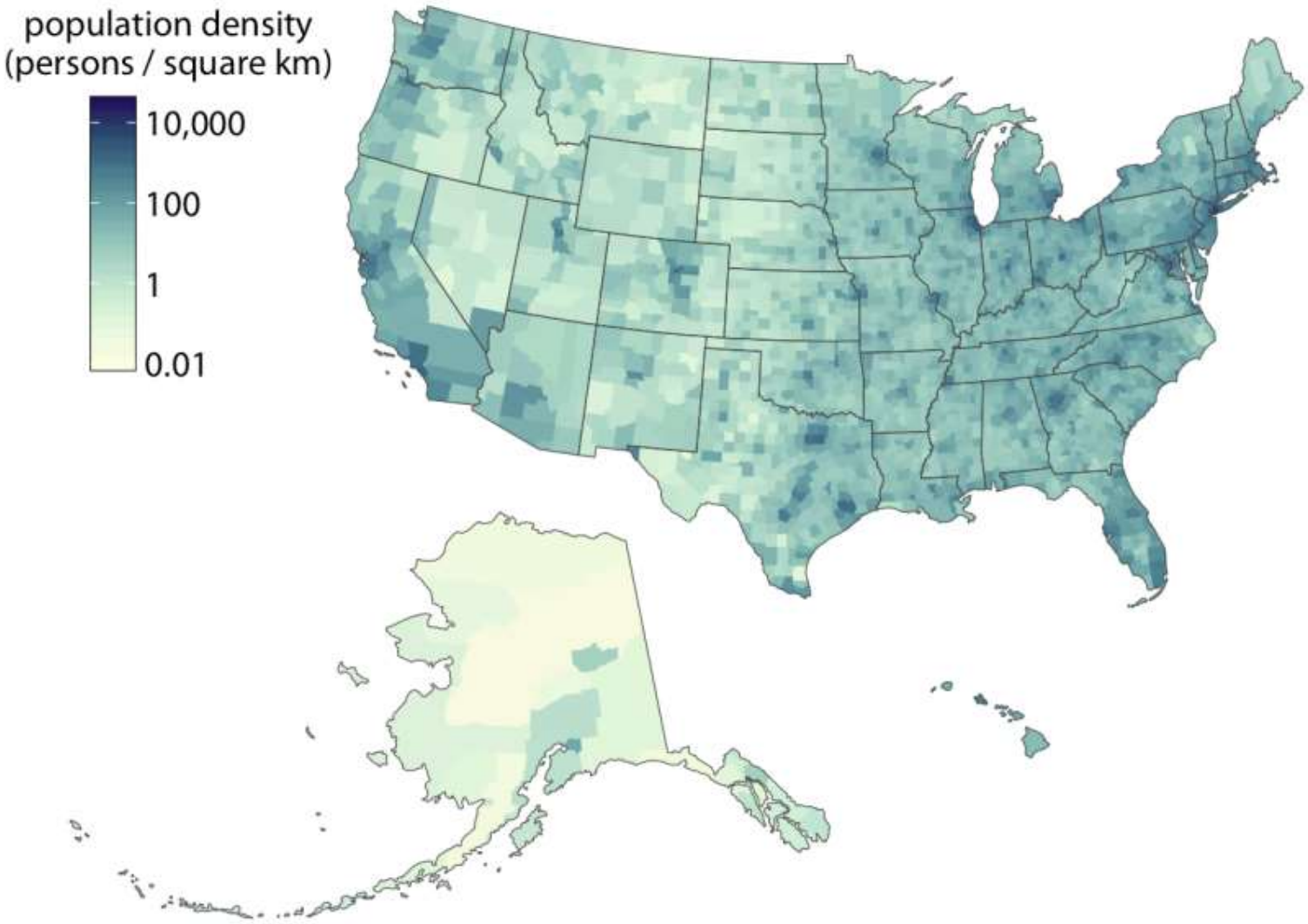
- To see locations in the physical world
  - Map layers (corresponding projection)
  - Choropleth maps
  - Cartogram

MAP LAYERS



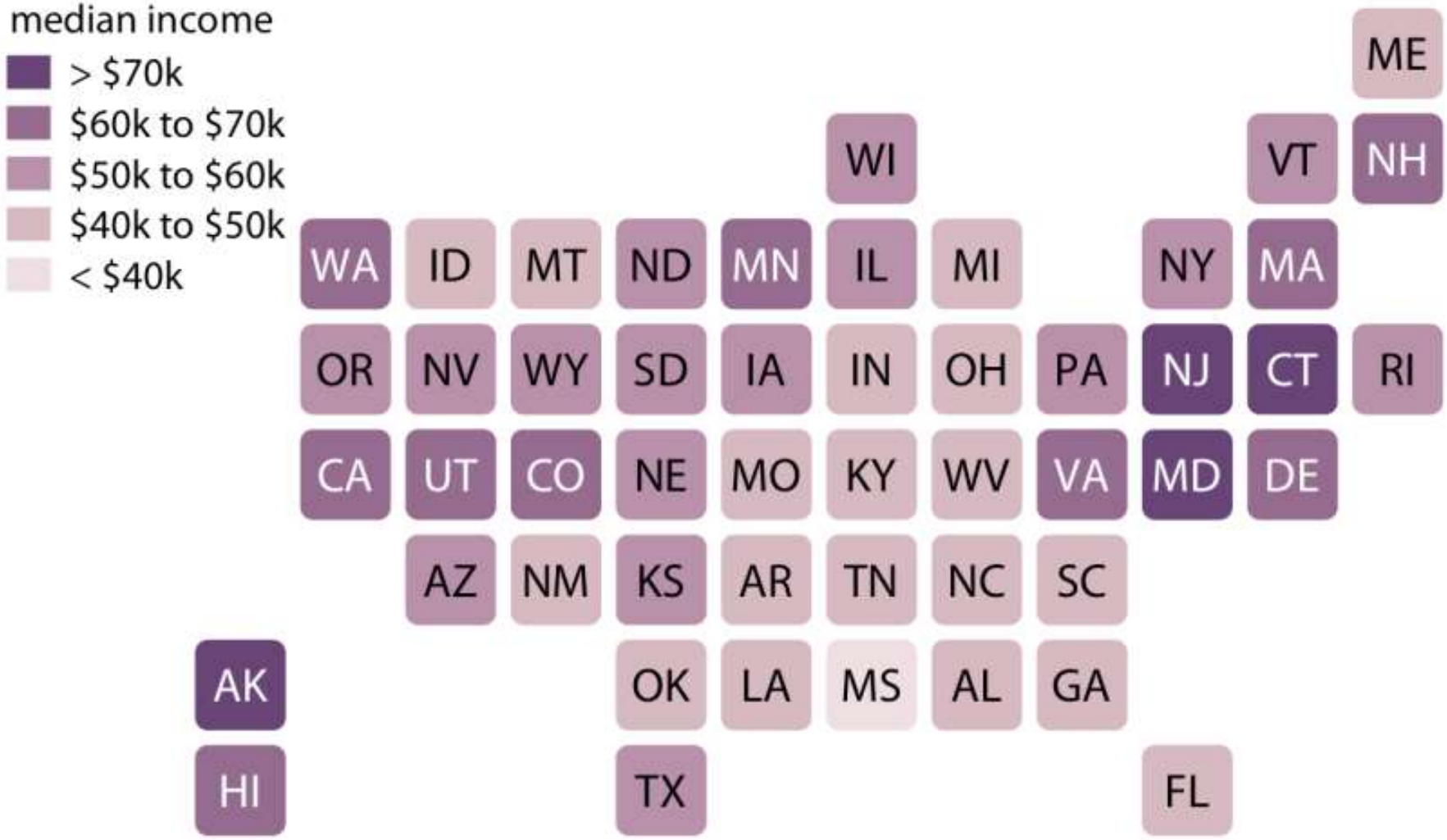
Claus O. Wilke, Fundamentals of data visualization: a primer on making informative and compelling figures, O'Reilly, 2019

CHOROPLETH MAPS



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CARTOGRAM

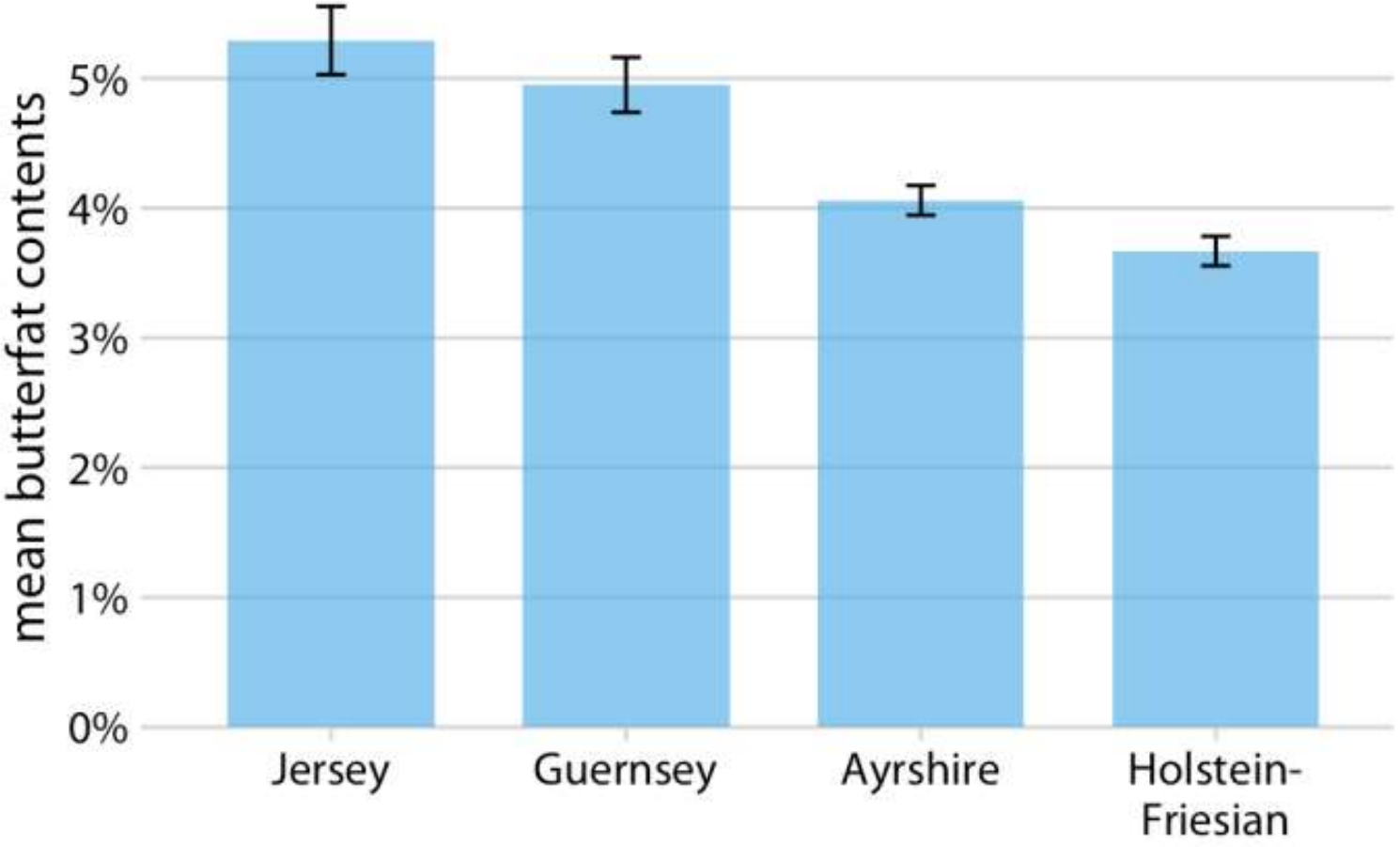


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VISUALIZING UNCERTAINTY

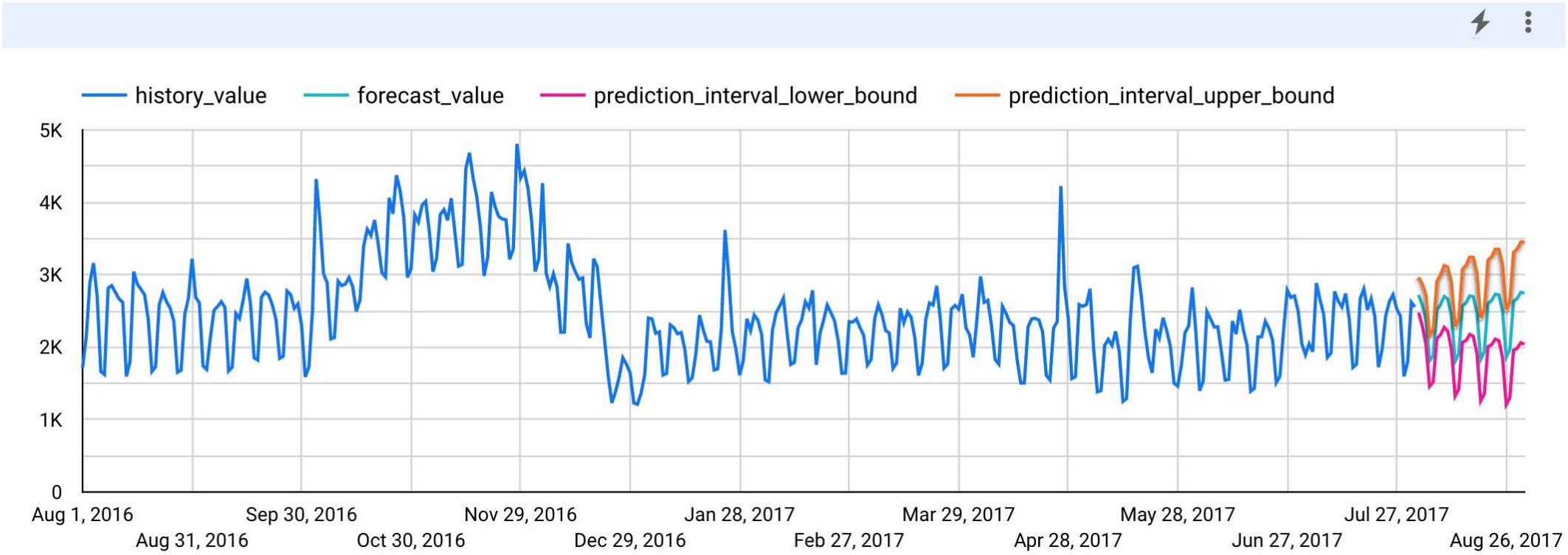
- Data with uncertainty
  - Error bars
  - Confidence bands

ERROR BARS



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CONFIDENCE BANDS



https://cloud.google.com/bigquery-ml/docs/arima-single-time-series-forecasting-tutorial

SUMMARY

- Business Intelligence Systems
- Fundamental of Visualization

QUESTIONS AND ANSWERS



Picture from: http://philadelphiasculpturegym.blogspot.com/2013/09/save-date-free-talk-and-q-on-affordable.html

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8. Ramesh Sharda, Dursun Delen, Efraim Turban, “Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support,” 7th Edition, Pearson Education, Inc., 2020.